CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

9.3 Resistance & Resistivity

Syllabus Section: 9. Electricity **Paper Type:** Multiple Choice (Paper 1) **Total:** 60 Questions (60 Marks)

Question 1 Easy [1 mark]

Which equation is used to define resistance?

A energy = (current)^2 × resistance × time

B potential difference = current × resistance

C power = (current)^2 × resistance

D resistivity = resistance × area ÷ length

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Resistance is defined by Ohm's Law, which states:
- The potential difference across a component is proportional to the current through it provided environmental conditions (e.g. temperature and pressure) stay constant
    - In other words, $V \quad \alpha \quad I$
- From this proportionality, we can say $V = kI$, where *k* is a constant
- This constant is what is known as resistance *R*
    - So, $V = IR$is the definition of resistance

Question 2 Easy [1 mark]

What is the unit of resistance when expressed in SI base units?

A kg m^2 s^{-2} A^{-1}

B kg m^2 s^{-3} A^{-2}

C kg m s^{-2} A^{-1}

D kg m s^{-3} A^{-1}

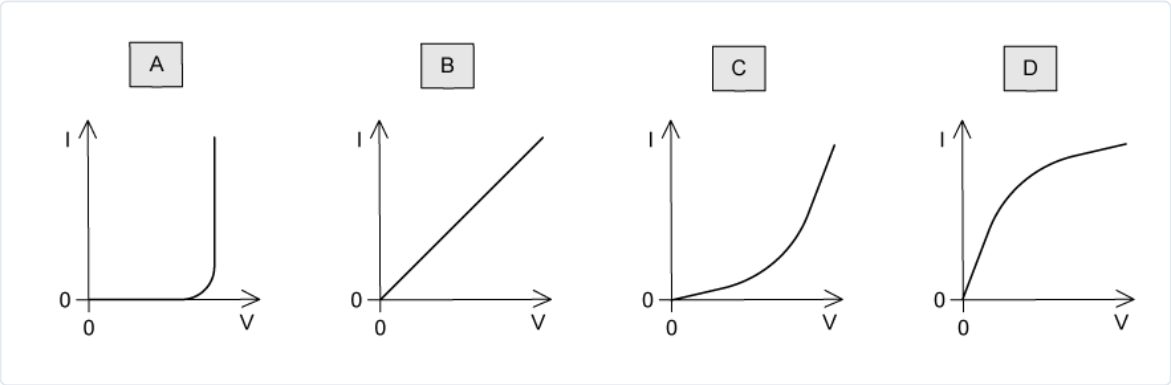
Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Resistance is defined as the ratio of voltage *V* across a component to the current *I* through it
  - In equation form, this can be expressed as $R = \frac{V}{I}$
- Potential difference is defined as the amount of electrical energy transferred between two points per unit of charge
  - In equation form, this can be expressed as $V = \frac{E}{Q}$
- The SI base units of potential difference are
  - $V = \frac{E}{Q} = \frac{\frac{1}{2}mv^2}{It} = kg \quad m^2 \quad s^{-2}A \quad s^{-1} = \text{kg} \quad \text{m}^2\,\text{s}^{-3}\,\text{A}^{-1}$
- So, the SI base units of resistance are
  - $R = \frac{V}{I} = kg \quad m^2 \quad s^{-3} \quad A^{-1}A^{-1} = \text{kg} \quad \text{m}^2 \quad \text{s}^{-3}\,\text{A}^{-2}$

Question 3 Easy [1 mark]

Which graph shows the *I–V* characteristic of a filament lamp ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The resistance of a filament lamp increases with temperature so the ratio $\frac{V}{I}$ increases as current increases
- As a result, the gradient decreases and a greater change in the potential difference is required to cause a change in the current
- On the *I–V* graph, it can be seen as the gradient on the curve becomes shallower, indicating a greater resistance
- The reason for this is that as the potential difference across a filament lamp increases, the current increases and the energy dissipated, as heat, increases, resulting in a higher temperature
- As the temperature increases, the resistance of the filament increases and the number of collisions between the free electrons and the lattice ions increases due to more rigorous vibration of the lattice ions

A is incorrect because this is the *I–V* graph for a semiconductor diode

B is incorrect because this is the *I–V* graph for an Ohmic conductor (obeys Ohm's Law)

C is incorrect because this is the *I–V* graph for a thermistor

Question 4 Easy [1 mark]

The resistance *R* of a resistor is determined by measuring the potential difference *V* across it and the current *I* in it. The value of *R* is then calculated using the equation

$$R = \frac{V}{I}$$

The values measured are *V* = 1.00 ± 0.05 V and *I* = 0.50 ± 0.01 A.

What is the percentage uncertainty in the value of *R* ?

A 2.5 %

B 3.0 %

C 7.0 %

D 10.0 %

Mark Scheme & Explanation:

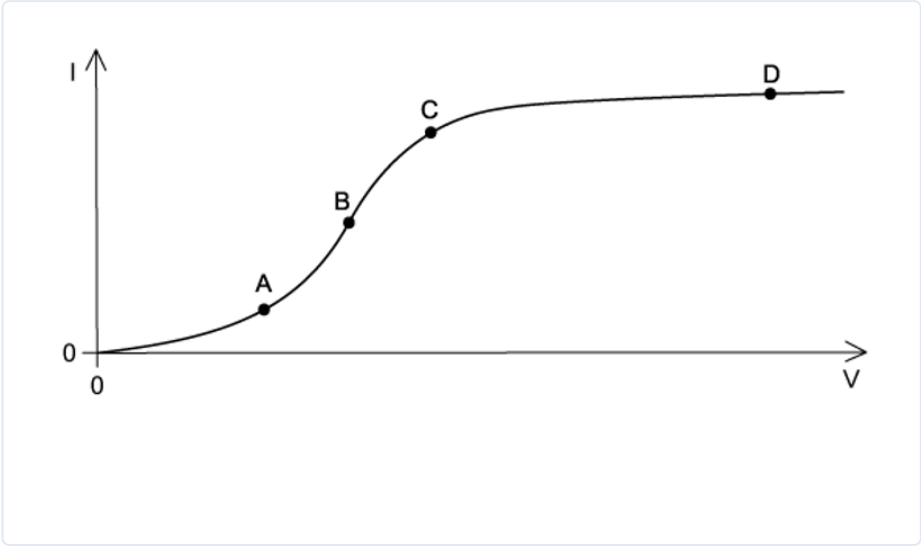
The correct answer is ***C*** *because:*

- From the equation given, the formula to work out the uncertainty in resistance is
  - $\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$
- To work out uncertainty in data which is multiplied or divided we add the percentage uncertainties
  - $\%\ \ uncertainity = \frac{uncertainity}{value} \times 100$
  - $\frac{\Delta R}{R} \times 100 = \frac{0.05}{1.00} \times 100 + \frac{0.01}{0.50} \times 100 = 5.0\ \ \% + 2.0\ \ \% = 7.0\ \ \%$

Question 5 Easy [1 mark]

The graph shows how the electric current *I* through a conducting liquid varies with the potential difference *V* across it.

At which point on the graph does the liquid have the smallest resistance?



A

B

C

D

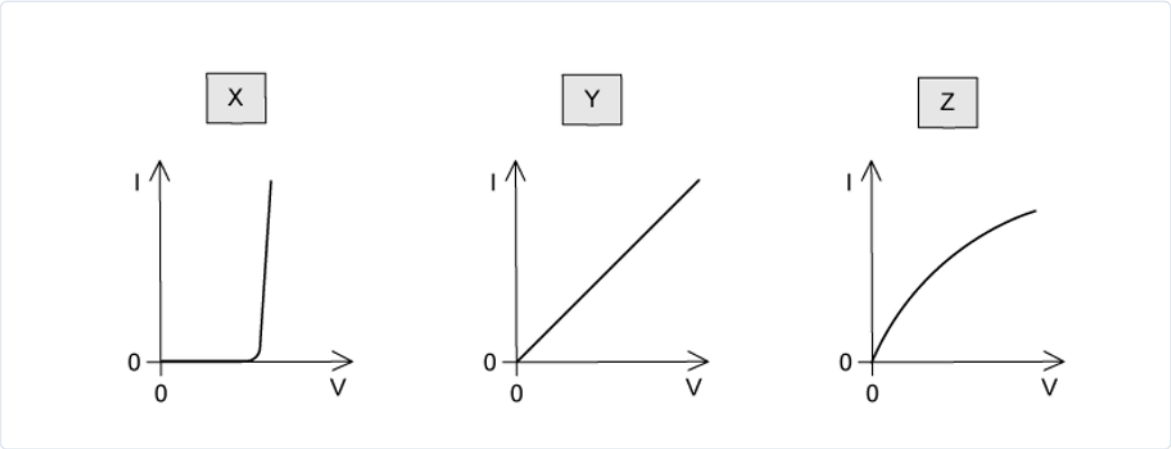
Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Ohm's Law is defined by the equation $V = IR$
- Resistance, however, is the potential difference per unit current $R = \frac{V}{I}$
- In relation to the *I–V* graph, we are looking for the point on the curve which gives us the maximum gradient
  - This is because the gradient is equal to $\frac{\Delta y}{\Delta x} = \frac{I}{V} = \frac{1}{R}$ so a maximum gradient will give us the smallest *R*
- The point with the maximum gradient is point **B**

Question 6 Easy [1 mark]

The graphs show the variation with potential difference *V* of the current *I* for three circuit components.



The components are a metal wire at constant temperature, a semiconductor diode and a filament lamp.

Which row of the table correctly identifies these graphs?

	Metal wire at constant temperature	Semiconductor diode	Filament lamp
A	X	Z	Y
B	Y	X	Z
C	Y	Z	X
D	Z	X	Y

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Graph X describes a conductor which has a very low resistance in one direction (low $\frac{V}{I}$) and a very high resistance in the other direction (high $\frac{V}{I}$)
    - This matches the description of a semiconductor diode
- Graph Y describes a conductor which obeys Ohm's Law, which means:

- The potential difference across a component is proportional to the current through it provided physical conditions (e.g. temperature and pressure) stay constant – in other words, $V \quad \alpha \quad I$
    - This matches the description of a metal wire at constant temperature
- Graph Z describes a conductor in which the resistance increases due to a change in environmental conditions (e.g. temperature) as the ratio $\frac{V}{I}$ increases
    - This matches the description of a filament lamp

Question 7 Easy [1 mark]

What is a correct statement of Ohm's Law?

A the potential difference across a component equals the current providing the resistance and other physical conditions stay constant

B the potential difference across a component equals the current multiplied by the resistance

C the potential difference across a component is proportional to its resistance.

D the potential difference across a component is proportional to the current in it providing physical conditions stay constant

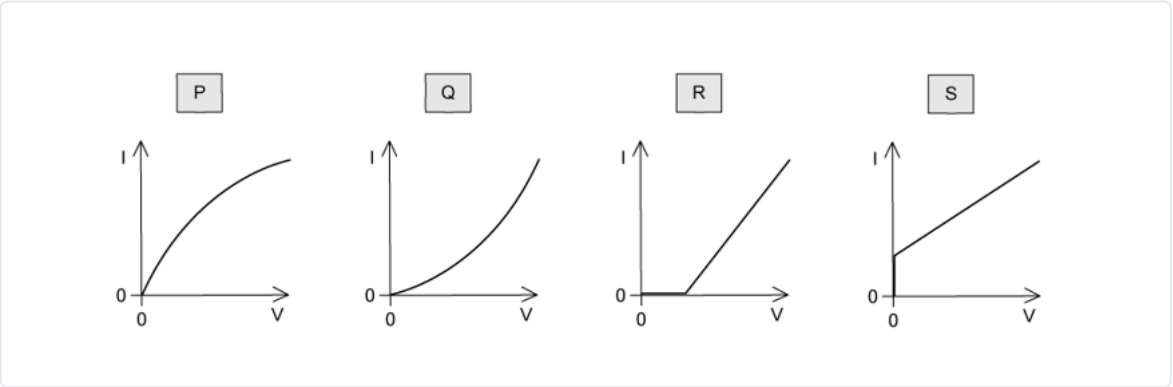
Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Resistance is defined by Ohm's Law, which states:
- The potential difference across a component is proportional to the current through it provided physical conditions (e.g. temperature and pressure) stay constant
    - In other words, $V \quad \alpha \quad I$
- From this proportionality, we can say $V = kI$, where *k* is a constant
- This constant is what is known as resistance *R*
    - So, $V = IR$ is the definition of resistance

Question 8 Easy [1 mark]

The graphs show possible current-voltage (*I*–*V*) relationships for a filament lamp and a semiconductor diode.



Which row best specifies the correct *I*–*V* graphs for the lamp and the diode?

	Filament lamp	Semiconductor diode
A	P	R
B	P	S
C	Q	R
D	Q	S

A

B

C

D

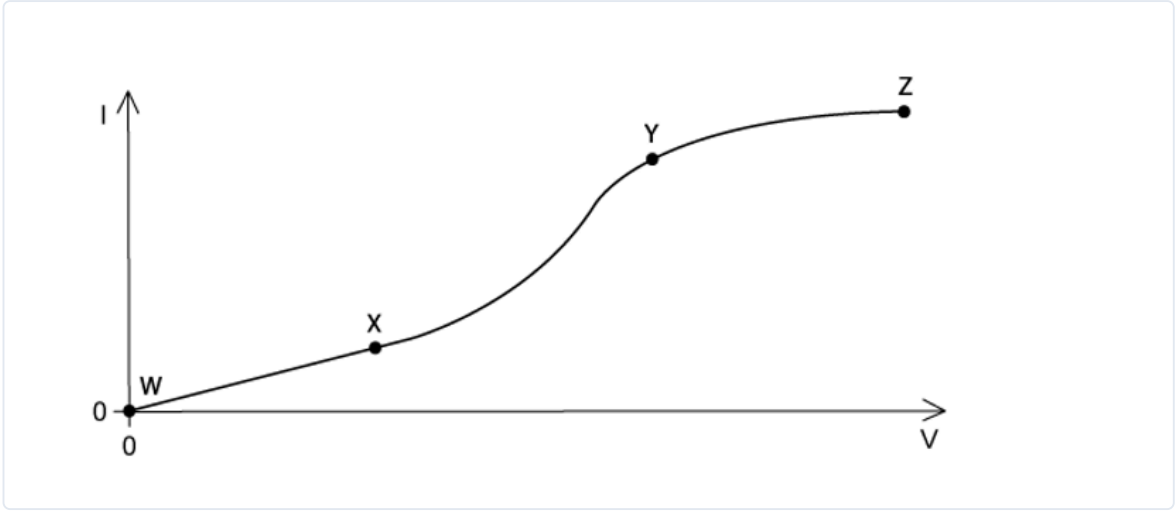
Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Graph P describes a conductor in which the resistance increases due to a change in environmental conditions (e.g. temperature) as the ratio $\frac{V}{I}$ increases
  - This matches the description of a filament lamp
- Graph R describes a conductor which has a very low resistance in one direction (low $\frac{V}{I}$) and a very high resistance in the other direction (high $\frac{V}{I}$)
  - This matches the description of a semiconductor diode

Question 9 Easy [1 mark]

An electrical component has a potential difference *V* across it and a current *I* through it. A graph of *I* against *V* is drawn and is marked in three sections WX, XY and YZ.



In which ways does the resistance of the component vary within each of the three sections?

	WX	XY	YZ
A	constant	decreases	increases
B	constant	increases	increases
C	increases	decreases	constant
D	increases	increases	decreases

A

B

C

D

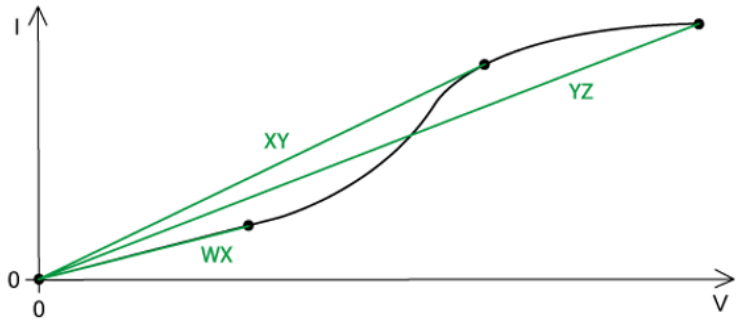
Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Resistance is defined as the potential difference per unit current $R = \frac{V}{I}$
- In section WX, the straight line indicates $V \quad \alpha \quad I$, so *R* is constant here
- In section XY, the current is increasing at a higher rate than the potential difference, which leads to a decrease in *R*

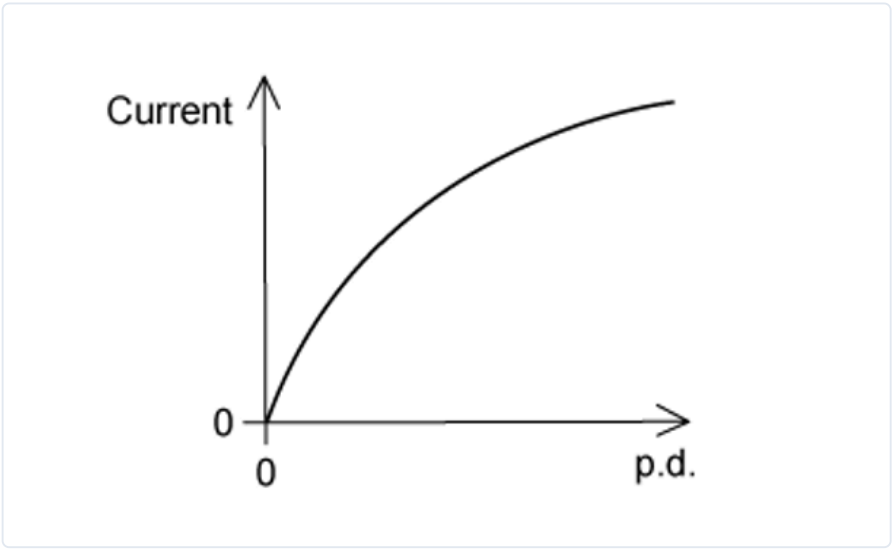
- As well as this, the gradient becomes steeper, meaning $\frac{I}{V} = \frac{1}{R}$ is increasing, which leads to a decrease in *R*

- In section YZ, the potential difference is increasing at a higher rate than the current, which leads to an increase in *R*
  - As well as this, the gradient becomes less steep, meaning $\frac{I}{V} = \frac{1}{R}$ is decreasing, which leads to an increase in *R*



Question 10 Easy [1 mark]

The graph shows the variation with potential difference (p. d.) of the current in a lamp filament.



Which statement explains the shape of this graph?

A as the filament temperature rises, electrons can pass more easily through the filament

B it takes time for the filament to reach its working temperature

C the power output of the filament is proportional to the square of the current in it

D the resistance of the filament increases with a rise in temperature

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Ohm's Law is defined by the equation $V = IR$
- At any point on the graph, the ratio of gives the value of $\frac{I}{V}$ where R is the resistance of $\frac{1}{R}$ the filament lamp
- It can be seen from the graph that this ratio decreases as the p. d. increases since the gradient decreases
- So, this means that the resistance R of the filament lamp increases
- When resistance increases, less current can flow
- Note that when current passes through the filament, its temperature increases
- Therefore, from the graph, it can be concluded that the resistance of the filament increases with a rise in temperature

A is incorrect because as the filament temperature rises, the resistance increases, so electrons pass through the filament **less** easily

B & C are incorrect because these options do not explain the shape of the graph

Question 11 Easy [1 mark]

The unit of resistivity, expressed in terms of base units, is given by

$$\text{kg} \quad x^3 \quad y^{-2} \quad z^{-3}$$

Which base units are *x*, *y* and *z* ?

	x	*y*	*z*
A	ampere	metre	second
B	metre	ampere	second
C	metre	second	ampere
D	second	ampere	metre

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Resistance and resistivity are related by the equation $R = \frac{\rho L}{A}$
- So, resistivity is equal to $\rho = \frac{RA}{L}$
- Ohm's law $V = IR$ and electrical power $P = VI$ can combine to make
  - $P = IRI = I^2R$
- So, resistance is equal to $R = \frac{P}{I^2}$
- Power is equal to $\frac{energy}{time}$
  - So, the units of power are $P = \frac{\frac{1}{2}mv^2}{t} = \text{kg} \quad \text{m}^2 \quad \text{s}^{-2}\,\text{s}^{-1} = \text{kg} \quad \text{m}^2 \quad \text{s}^{-3}$
  - Therefore, the units of resistance are $R = \frac{P}{I^2} = \text{kg} \quad \text{m}^2 \quad \text{s}^{-3}\,\text{A}^{-2} = \text{kg} \quad \text{m}^2 \quad \text{s}^{-3} \quad \text{A}^{-2}$
- And hence, the units of resistivity are
  - $\rho = \frac{RA}{L} = \text{kg} \quad \text{m}^2\,\text{s}^{-3}\,\text{A}^{-2}\,\text{m}^2\,\text{m}^{-1} = \text{kg} \quad \text{m}^3\,\text{s}^{-3}\,\text{A}^{-2}$
- If we compare this to $\text{kg} \quad x^3 y^{-2} \quad z^{-3}$, we can see ***x* is meters, *y* is amperes, and *z* is seconds**

Question 12 Easy [1 mark]

A cylindrical wire of length 10 m and diameter 2.0 mm has a resistance of 0.050 Ω.

From which material is the wire made?

	Material	Resistivity / Ω m
A	bronze	1.6×10^{-7}
B	nichrome	1.6×10^{-6}
C	silver	1.6×10^{-8}
D	zinc	6.3×10^{-8}

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Resistance and resistivity are related by the equation $R = \frac{\rho L}{A}$
- So, resistivity is equal to $\rho = \frac{RA}{L}$
- The wire is cylindrical, so the cross-sectional area is equal to
  - $A = \pi r^2 = \pi \frac{d}{2}^2 = \pi \frac{2.0 \times 10^{-3}}{2}^2$
- Therefore, the resistivity is equal to
  - $\rho = \frac{0.050}{10} \times \pi \frac{2.0 \times 10^{-3}}{2}^2 = 1.57 \times 10^{-8}\ \Omega\,\text{m}$
- Therefore, the wire is made from **silver**, which has a resistivity of $6 \times 10^{-8}\ \Omega\,\text{m}$

Question 13 Easy [1 mark]

The resistance of one device R_1 is designed to change with temperature; the resistance of another device R_2 is designed to change with illumination.

What are the names of these devices ?

	R_1	R_2
A	thermistor	light-dependent resistor
B	thermistor	potential divider
C	semiconductor diode	light-dependent resistor
D	semiconductor diode	potential divider

A

B

C

D

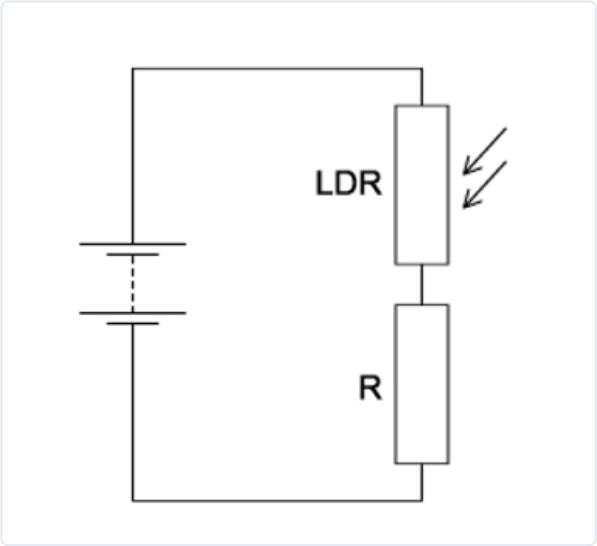
Mark Scheme & Explanation:

*The correct answer is **A** because:*

- A device whose resistance depends on temperature is called a **thermistor**
  - For a thermistor with a negative temperature coefficient, the resistance decreases significantly with a rise in temperature
- A device whose resistance depends on illumination is called a **light-dependent resistor or LDR**
  - The higher the intensity of light on the LDR, the greater the number of electrons in the semiconducting material that can move freely. Hence, as the light intensity increases, the resistance of the LDR decrease

Question 14 Easy [1 mark]

A light-dependent resistor (LDR) is connected in series with a resistor *R* and a battery.



The resistance of the LDR is equal to the resistance of *R* when no light falls on the LDR. When the light intensity falling on the LDR increases, which statement is correct?

A the current in *R* decreases

B the current through the LDR decreases

C the p. d. across *R* decreases

D the p .d. across the LDR decreases

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The resistance of the LDR decreases as the light intensity increases
- Since the LDR and resistor R are connected in series, the current in both of them is the same
- Ohm's law states $V = IR$
- Since the resistance of the LDR decreases, the potential difference V across it also decreases

A & **B** are incorrect because a decrease in the resistance of the LDR causes the total resistance R in the circuit to decrease, from Ohm's law $I = \frac{V}{R}$, we can see the current through both the LDR and resistor R increases

C is incorrect because the battery has a fixed e. m. f., so this causes the p. d. across R to increase

Question 15 Easy [1 mark]

A copper wire is stretched so that its diameter is reduced from 1.0 mm to a uniform 0.5 mm. The resistance of the unstretched copper wire is 0.2 Ω.

What is the resistance of the stretched wire?

A 0.4 Ω

B 0.8 Ω

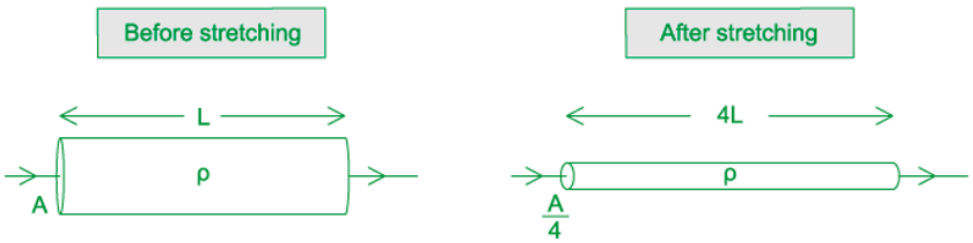
C 1.6 Ω

D 3.2 Ω

Mark Scheme & Explanation:

*The correct answer is **D** because:*

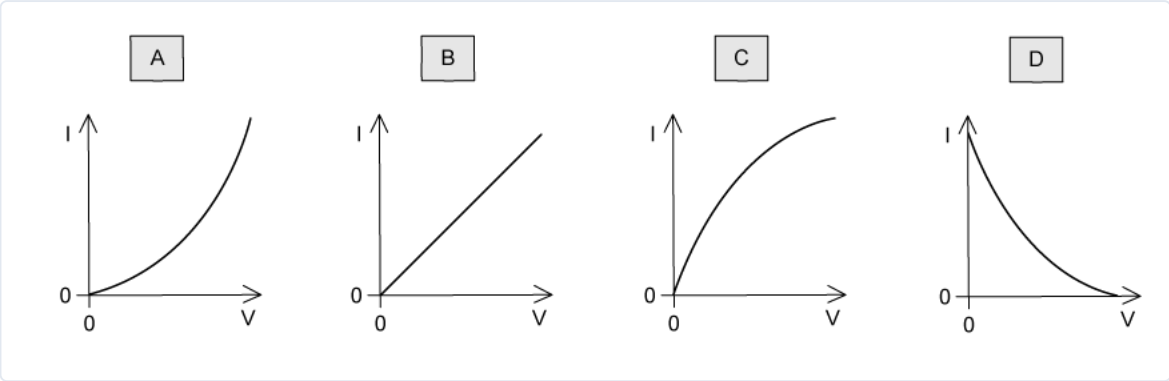
- Resistance R of a wire is equal to $R = \frac{\rho L}{A}$
- Cross-sectional area is equal to $A = \pi r^2 = \pi \frac{d}{2}^2 = \frac{\pi d^2}{4}$
- The resistance *R* is inversely proportional to the area, and therefore, *R* is also inversely proportional to the diameter squared $R \propto \frac{1}{d^2}$
  - So, when the wire is stretched, the diameter is reduced by half, which causes the area to be reduced by 4
  - So, the resistance *R* is increased by a factor of 4 due to the cross-sectional area *A*
- Since the volume of the wire is always constant, we can use this to find the new length – the volume of the unstretched wire is $V = AL$
  - Where *L* is the length of the wire
- The stretched wire has ¼ the area of the unstretched wire, for its volume to remain constant, its length must have been increased by a factor of 4
  - The volume of the stretched wire is $V = \frac{A}{4} 4L$
- So, the length of the stretched wire is 4*L*
  - Resistance *R* of unstretched wire $= \frac{\rho L}{A} = 0.2\ \Omega$
  - Resistance *R* of stretched wire $= \frac{\rho 4L}{\frac{1}{4}A} = 16\frac{\rho L}{A} = 160.2 = 3.2\ \Omega$



Volume remains constant i.e. $V = AL = \frac{A}{4} \times 4L$

Question 16 Easy [1 mark]

Which graph best represents the way in which the current *I* through a thermistor depends upon the potential difference *V* across it ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- As the potential difference *V* across the thermistor increases, the temperature of the thermistor also increases, causing its resistance *R* to decrease
    - Ohm's law states $V = IR$
- As the p.d. increases, the current increases - this is represented by a positive gradient in the graph
    - Resistance is equal to $R = \frac{V}{I}$ or $\frac{1}{R} = \frac{I}{V}$
- As the resistance *R* decreases, the value of $\frac{1}{R}$ increases, so, the ratio $\frac{V}{I}$ increases
- So, current *I* increases with the p .d., with an increasing gradient since the resistance *R* is also decreasing as the p. d. increases
    - The graph that matches this description is option A

B is incorrect because the resistance of the thermistor decreases as p .d. increases, so it does not obey Ohm's Law

C is incorrect because the resistance of the thermistor decreases as p. d. increases, whereas this graph shows the resistance decreasing

D is incorrect because when the p. d. increases, the current increases so the graph should have a positive gradient

Exam Tip: It is important to note, that the gradient only tells us the way the resistance is changing (increasing or decreasing). We cannot use the value of the gradient to obtain the value of the resistance. Instead, we would need to consider the ratio of $\frac{V}{I}$ of a single point.

Question 17 Easy [1 mark]

A power cable has length 2000 m. The cable is made of twelve parallel strands of copper wire, each with diameter 0.51 mm.

What is the resistance of the cable ? (resistivity of copper = 1.7×10^{-8} Ω m)

A 0.014 Ω

B 3.5 Ω

C 14 Ω

D 166 Ω

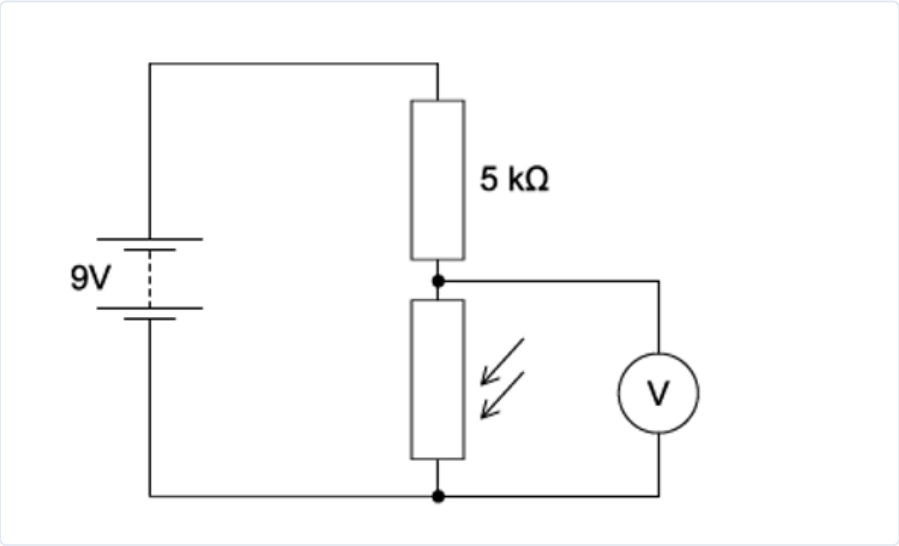
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Resistance *R* of a wire is equal to $R = \frac{\rho L}{A}$
- Cross-sectional area is equal to $A = \pi r^2 = \pi \frac{d}{2}^2 = \frac{\pi d^2}{4}$
- Cross-sectional area of one strand is equal to $= \frac{\pi 0.51 \times 10^{-3}{}^2}{4} = 2.0 \times 10^{-7} \text{ m}^2$
- Total cross-section of cable $= 12 \times 2.0 \times 10^{-7} = 2.45 \times 10^{-6} \text{ m}^2$
  - $R = \frac{\rho L}{A} = \frac{1.7 \times 10^{-8} 2000}{2.45 \times 10^{-6}} = 13.87 \ \Omega$
- Therefore, the resistance of the cable is **14 Ω**

Question 18 Easy [1 mark]

A circuit is set up with an LDR and a fixed resistor as shown.



The voltmeter reads 4 V.

The light intensity is increased.

What is a possible voltmeter reading?

A 3V

B 4V

C 6V

D 8V

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The resistance of an LDR decreases as the light intensity incident on it increases
- Ohm's law states or $V = IR$ or $R = \frac{V}{I}$
- The p. d. across a component is dependent on its resistance
- Since the resistance of the LDR decreases, the p. d. across it will also decrease

- Therefore, if the original p. d. is 4 V, the only possible reading it can be when the light intensity is increased is 3 V

Question 19 Easy [1 mark]

A cylindrical wire 4.0 m long has a resistance of 31 Ω and is made of metal of resistivity 1.0×10^{-6} Ω m.

What is the radius of cross-section of the wire?

A 1.0×10^{-8} m

B 2.0×10^{-8} m

C 6.4×10^{-8} m

D 2.0×10^{-4} m

Mark Scheme & Explanation:

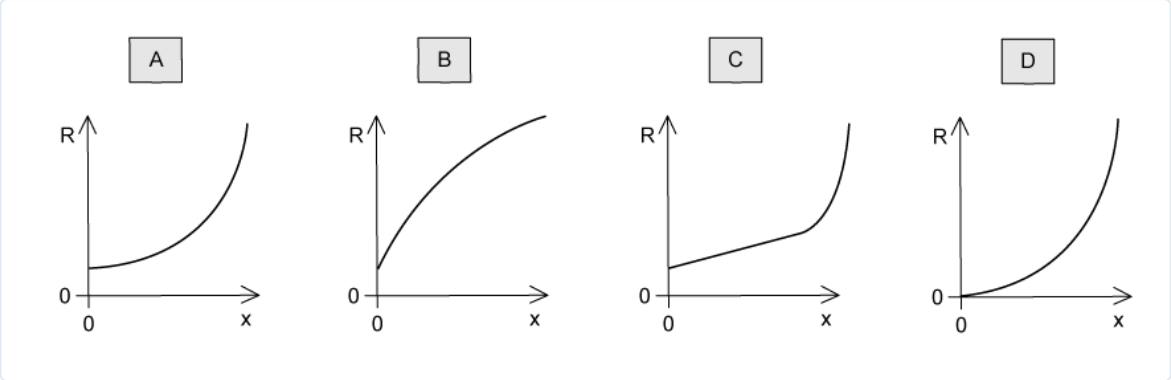
*The correct answer is **D** because:*

- Resistance *R* of a wire is equal to $R = \frac{\rho L}{A}$
  - $A = \frac{\rho L}{R} = \frac{1.0 \times 10^{-6} 4.0}{31} = 1.29 \times 10^{-7} \quad \text{m}^2$
- Cross-sectional area is equal to $A = \pi r^2$, so, $r = \sqrt{\frac{A}{\pi}}$
  - So, the radius is $= \sqrt{\frac{1.29 \times 10^{-7}}{\pi}} = 2.0 \times 10^{-4} \quad m$

Question 20 Easy [1 mark]

When a thin metal wire is stretched, it becomes longer and thinner. This causes a change in the resistance of the wire. The volume of the wire remains constant.

Which graph could represent the variation with extension *x* of the resistance *R* of the wire ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- If you stretch a wire to double its length, its cross-sectional area is halved
- This is because its volume must stay the same, i.e.:
  - If the volume of the unstretched wire is $V = AL$
  - Then, the volume of the stretched wire is $V = \frac{A}{2}2L$
- So, in the above formula, you are multiplying L by 2 and dividing A by 2, this means the resistance goes up by a factor of 4
  - $\frac{\rho 2L}{\frac{1}{2}A} = 4\frac{\rho L}{A} = 4R$
- Therefore, the relationship between extension and resistance is a square law
  - $R \ \alpha \ x^{\ 2}$
- At zero extension, there will be some resistance, so, the graph does not start from the origin
  - Therefore, the graph that best fits this description is option A

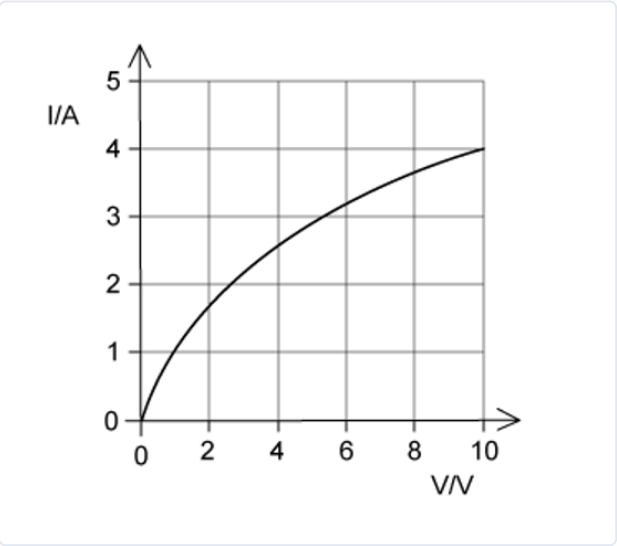
B is incorrect because even though this graph shows a non-linear relationship, it shows an initial sharp rise in resistance which levels off and becomes constant, but this is not the case - the resistance will continue to rise with extension

C is incorrect because this graph shows the first section as a linear relationship which is incorrect

D is incorrect because this graph shows the resistance starts from the origin, but at zero extension, there will be some resistance so this is incorrect

Question 21 Medium [1 mark]

The graph shows how current *I* varies with voltage *V* for a filament lamp.



Since the graph is not a straight line, the resistance of the lamp varies with *V*.

Which row gives the correct resistance at the stated value of *V*?

	***V* / V**	***R* / Ω**
A	2.0	1.5
B	4.0	3.2
C	6.0	1.9
D	8.0	0.9

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Ohm's Law is defined by the equation $V = IR$
- A filament lamp does not follow Ohm's Law since the variation of I with V is not linear, and a straight line is not obtained

- Resistance is equal to $R = \frac{V}{I}$
- Consider option **C**:
    - At *V* = 6.0 V, current *I* = 3.2 A
    - Resistance $R = \frac{V}{I} = \frac{6.0}{3.2} = 1.875\Omega$
- Therefore, the resistance at 6.0 V is 1.9 Ω; this matches option **C** correctly

A is incorrect because At *V* = 2.0 V, current *I* = 1.8 A, Resistance $R = \frac{2.0}{1.8} = 1.11\Omega$

This value ≠ 1.5 Ω

B is incorrect because At *V* = 4.0 V, current *I* = 2.5 A, Resistance $R = \frac{4.0}{2.5} = 1.6\ \Omega$

This value ≠ 3.2 Ω

D is incorrect because At *V* = 8.0 V, current *I* = 3.6 A, Resistance $R = \frac{8.0}{3.6} = 2.2\ \Omega$

This value ≠ 0.9 Ω

Exam Tip: The values of current *I* read from the graph may not be exact for all the points, but they are close to the correct value. You do not need to be extra precise in these types of question (but you do in the written answer questions!)

Question 22 Medium [1 mark]

A charge of 8.0 C passes through a resistor of resistance 30 Ω at a constant rate in a time of 20 s.

What is the potential difference across the resistor?

A 0.40 V

B 5.3 V

C 12 V

D 75 V

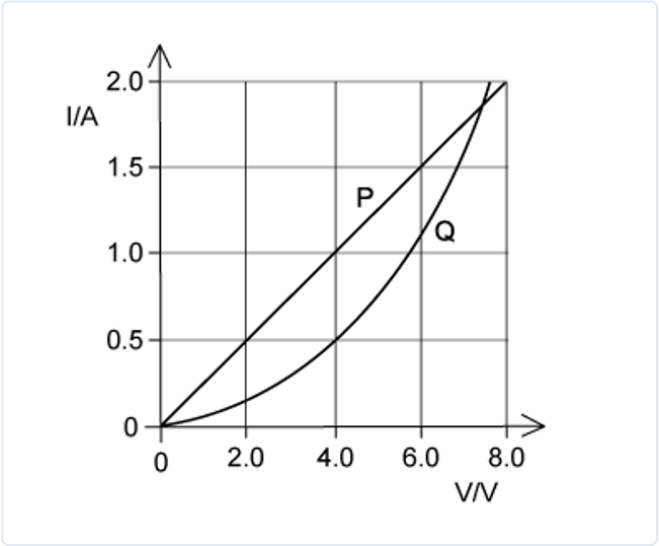
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Potential difference across a resistor is equal to $V = IR$
- Current is equal to $I = \frac{Q}{t}$
- Combining these equations gives $V = \frac{QR}{t}$
    - So, the potential difference is $V = \frac{8.0 \times 3.0}{20} = 12 \quad \text{V}$

Question 23 Medium [1 mark]

The *I* -*V* characteristics of two electrical components P and Q are shown below.



Which statement is correct ?

A P is a resistor and Q is a filament lamp

B the resistance of Q increases as the current in it increases

C for a current of 1.9 A, the resistance of Q is approximately half that of P

D for a current of 0.5 A, the power dissipated in Q is double that in P

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Graph P shows a component which obeys Ohm's Law
- Graph Q shows a component whose resistance decreases with *V*, unlike the graph for a filament lamp
- Power dissipated is equal to $P = VI$
- At current *I* = 0.5 A
  - Graph P: *V* = 2.0 V
  - Graph Q: *V* = 4.0 V
- So, for graph Q, *V* is twice that of graph P ; therefore the power dissipated in Q is double that in P
  - Since $2VI = 2P$

A is incorrect because the *I*-*V* characteristics of a filament lamp would should a decreasing gradient, unlike the graph Q, which has an increasing gradient. This is because as *V* increases, the temperature of the lamp increases, causing the resistance to increase

B is incorrect because as *V* increases, the current *I* increases. From Ohm's Law, $I = \frac{V}{R}$, so the resistance *R* should be getting smaller

C is incorrect because at current *I* = 1.9 A, the 2 graphs intercept, so, they have the same value of *V* at that point, and hence the same resistance

Question 24 Medium [1 mark]

The resistance *R* of a resistor is to be determined. The current *I* in the resistor and the potential difference *V* across it are measured.

The results, with their uncertainties, are

I = (2.0 ± 0.2) A *V* = (15.0 ± 0.5) V

What is the uncertainty in this value for *R* ?

A ± 0.3 Ω

B ± 0.5 Ω

C ± 0.7 Ω

D ± 1 Ω

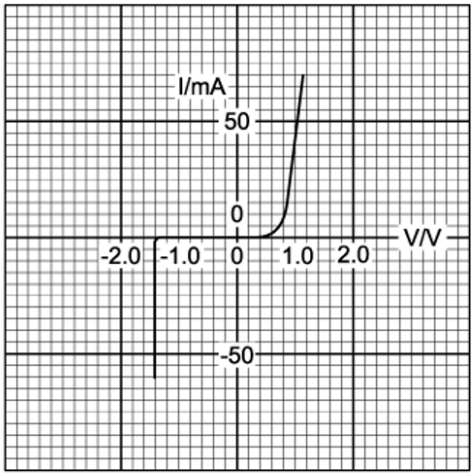
Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The value of *R* is $R = \frac{V}{I} = \frac{15.0}{2.0} = 7.5\Omega$
- From the equation , the formula to work out the uncertainty in resistance is
  - $\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$
- To work out uncertainty in data which is multiplied or divided we add the uncertainties
  - $\frac{\Delta R}{R} = \frac{0.5}{15.0} + \frac{0.2}{2.0} = \frac{2}{15}$
- Then, to find the uncertainty in *R* we multiply by the calculated value of *R*
  - $\Delta R = \frac{2}{15} \times 7.5 = \pm 1\ \Omega$

Question 25 Medium [1 mark]

The variation with potential difference *V* of the current *I* in a semiconductor diode is shown below.



What is the resistance of the diode for applied potential differences of +1.0 V and –1.0 V ?

	Resistance	
	at +1.0 V	**at –1.0 V**
A	20 Ω	infinite
B	20 Ω	zero
C	0.05 Ω	infinite
D	0.05 Ω	zero

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

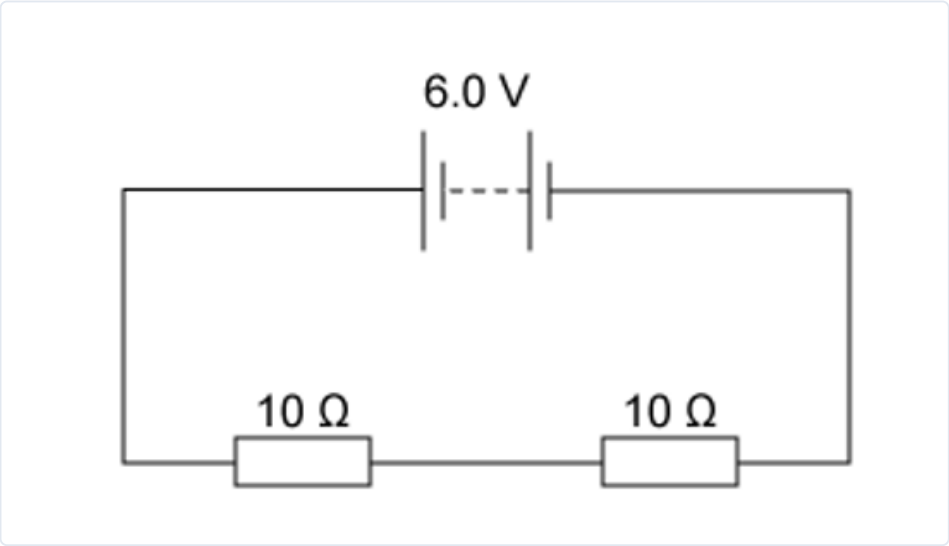
- A semiconductor diode is a device that has very low resistance in one direction and a very high resistance in the other direction

    - Think of it like a ball at the top of a hill: it can roll from top to bottom easily, but requires external energy to take it from the bottom to the top of the hill

- When there is no current, the resistance is infinite. Because of this, the electrical resistance of the diode is very high or infinite for negative voltages until about –1.5 V
- The electrical resistance of the diode decreases from 0.5 V onwards
- At 1.0 V, the current is equal to 50 mA or 0.05 A
    - So, the resistance at this point is $R = \frac{V}{I} = \frac{1.0}{0.05} = 20\ \Omega$

Question 26 Medium [1 mark]

A battery of negligible internal resistance is connected to two 10 Ω resistors in series.



What charge flows through each of the 10 Ω resistors in 1 minute?

A 0.30 C

B 0.60 C

C 3.0 C

D 18 C

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Potential difference across a resistor is equal to $V = IR$
- Current is equal to $I = \frac{Q}{t}$
- Combining these equations gives $V = \frac{QR}{t}$
- Rearranging this for charge Q gives $Q = \frac{Vt}{R}$
    - The p.d. through each 10 Ω resistor is $V = 3.0 \text{ V}$
    - The time is equal to $t = 60 \text{ s}$

- So, $Q = \frac{Vt}{R} = \frac{3.0 \times 60}{10} = 18 \quad \text{C}$

Question 27 Medium [1 mark]

Two heating coils X and Y, of resistance R_X and R_Y respectively, deliver the same power when 12 V is applied across X and 6 V is applied across Y.

What is the ratio $\frac{R_X}{R_Y}$?

A $\frac{1}{4}$

B $\frac{1}{2}$

C 2

D 4

Mark Scheme & Explanation:

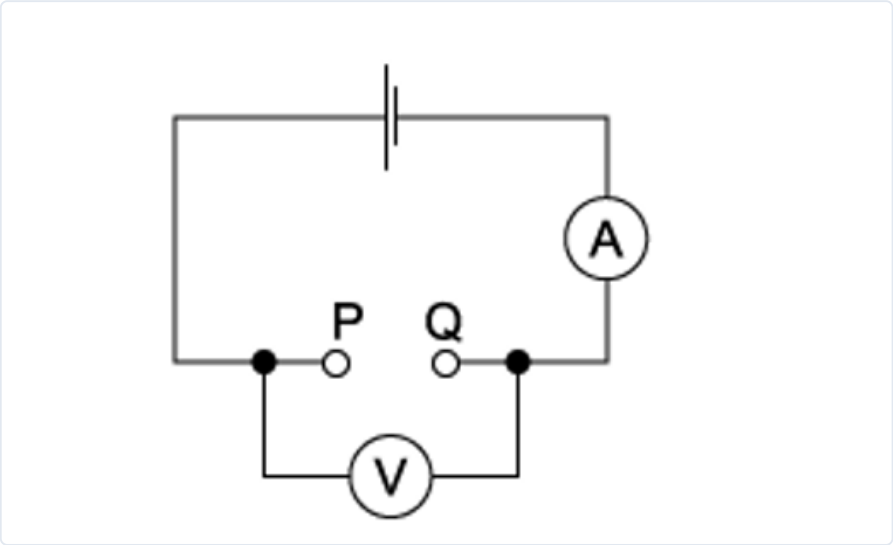
*The correct answer is **D** because:*

- Electrical power is defined as $P = VI$
- Ohm's Law is given by $V = IR$
- Substituting *I* for $I = \frac{V}{R}$
  - $P = V\frac{V}{R} = \frac{V^2}{R}$
- Power delivered by X = Power delivered by Y
  - $P_X = P_Y$
  - $\frac{12^2}{R_X} = \frac{6^2}{R_Y}$
  - $\frac{R_X}{R_Y} = \frac{144}{36} = 4$

Question 28 Medium [1 mark]

A student found two unmarked resistors. To determine the resistance of the resistors, the circuit below was set up. The resistors were connected in turn between P and Q, noting the current readings.

The voltage readings were noted without the resistors and with each resistor in turn.



The results were entered into a spreadsheet, as shown.

1.5	1.3	28	46
1.5	1.4	14	100

The student forgot to enter the column headings.

Which order of the headings would be correct?

A	e.m.f / V	*V* / V	*R* / Ω	*I* / mA
B	*V* / V	e.m.f / V	*R* / Ω	*I* / mA
C	*V* / V	e.m.f / V	*I* / mA	*R* / Ω

D	e.m.f / V	*V* / V	*I* / mA	*R* / Ω

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The e. m .f. of the supply in the circuit is constant and independent of the other components in the circuit. Without the resistors in the circuit, the voltmeter will read the e .m. f. of the supply. Since the e .m .f is constant, the first column, which is the only column with constant values, must be the e. m .f.
- The second column, which contain values close to 1.5, must be the p .d. across the resistor
- The values in this column increases from 1.3 to 1.4 when the values of the last column increase from 46 to 100
- Ohm's Law states $V = IR$, so as p. d. increases, *R* increases, so the final column must be *R*
- Additionally, it can be seen that when the resistance *R* is greater, the current is smaller

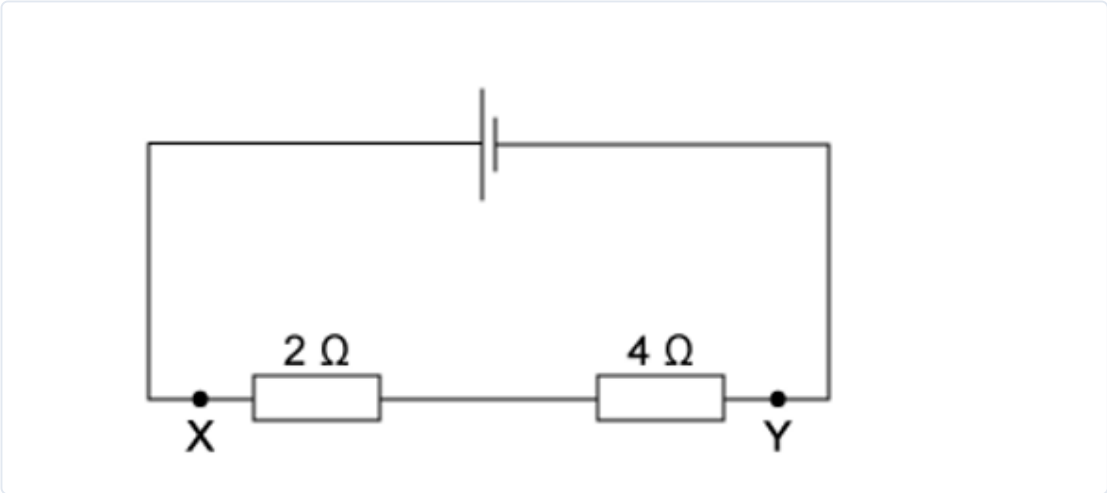
A is incorrect because if you consider the voltage drop due to internal resistance, equal to *Ir*, there would be a more significant drop when the current is greater, but this wouldn't make sense if the p. d. increased from 1.3 to 1.4 V if the current increased from 46 – 100 mA

B & C are incorrect because the first column must be e. m .f. as this value does not change

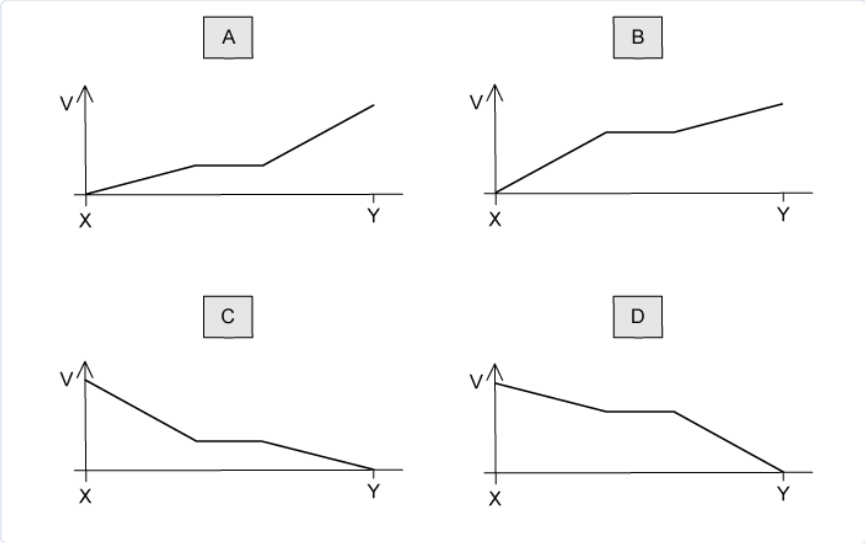
Exam Tip: Note that usually, when filling columns, as in this example, the first column would be the values that remain constant (if any – here it is the e .m .f.) and the last column is a quantity that needs to be calculated (here, the resistance). The middle columns usually contain the measured quantities (here, current and p .d.).

Question 29 Medium [1 mark]

A 2 Ω resistor and a 4 Ω resistor are connected to a cell.



Which graph shows how the potential V varies with distance between X and Y ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The cell has some e .m .f., say *E*. The negative terminal of the cell is usually taken to be at a potential of 0 V while the positive terminal is taken to be at a potential of *E*

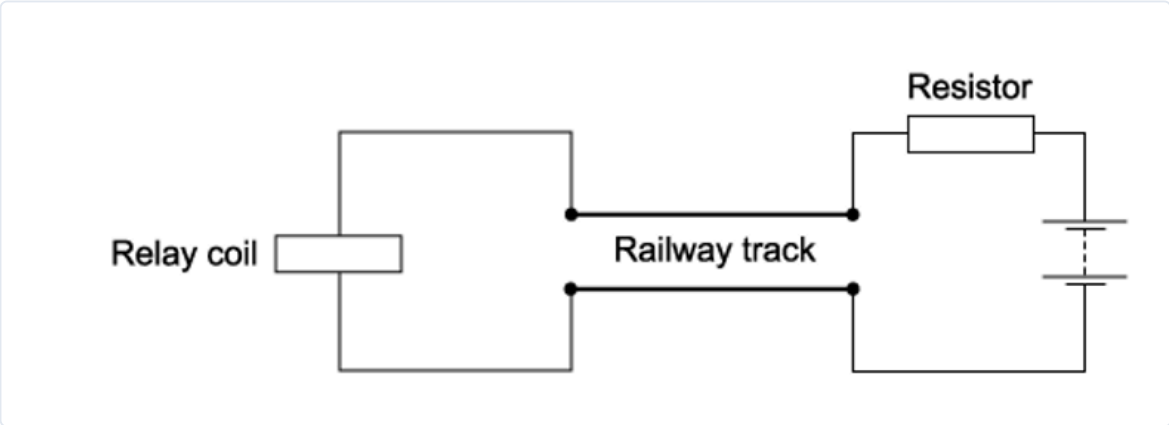
- So, the positive terminal is at a higher potential than the negative terminal
- No component is connected between the positive terminal and point X, so they are at the same potential
- Therefore, point X is at potential E, while, point Y is at the same potential as the negative terminal of the cell, so, the potential at Y is zero
- Ohm's Law states *V*= *IR*
- Therefore, the p. d. across the 2Ω resistor must be less than the p .d. across the 4Ω resistor. That is, the change in potential across the 4Ω resistor is more significant than that across the 2Ω resistor
- In the graph, this is represented by the gradient. The gradient across the 4Ω resistor, which is just before point Y, is steeper than that across the 2Ω resistor which is just after point X

A & B are incorrect because the potential at point X is *E* and point Y is zero so the gradient should decrease

C is incorrect because Ohm's Law tells us that $V \quad \alpha \quad R$, so the gradient is steeper across the 4Ω resistor and shallower across the 2Ω resistor

Question 30 Medium [1 mark]

The diagram shows a length of track from a model railway connected to a battery, a resistor and a relay coil.



With no train present, there is a current in the relay coil which operates a switch to turn on a light.

When a train occupies the section of track, most of the current flows through the wheels and axles of the train in preference to the relay coil. The switch in the relay turns off the light.

Why is a resistor placed between the battery and the track?

A to limit the heating of the wheels of the train

B to limit the energy lost in the relay coil when a train is present

C to prevent a short circuit of the battery when a train is present

D to protect the relay when a train is present

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- When the train is present, it is stated that most of the current flows through the wheels and axles of the train in preference to the relay coil
- So, there is little current in the relay coil; therefore, the energy lost in the relay coil when a train is present is already limited
- Ohm's Law states $V = IR$, so the link between current and resistance is $I = \frac{V}{R}$
- The resistor actually limits the current flowing in the circuit when the train is present
- Therefore, this prevents a short circuit of the battery when a train is present

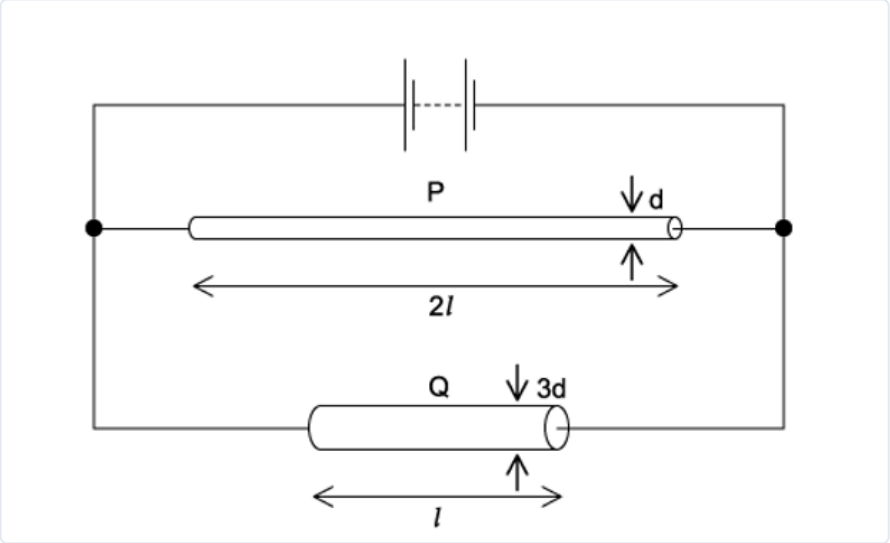
A is incorrect because the heating of the wheels due to the current is small compared to the heating due to its motion

B & D are incorrect because the relay is already protected, so this isn't the reason for needing a resistor

Question 31 Medium [1 mark]

Two wires P and Q made of the same material are connected to the same electrical supply.

P has twice the length of Q and one-third of the diameter of Q, as shown in the diagram.



What is the ratio $\frac{current\ \ in\ \ P}{current\ \ in\ \ Q}$?

A $\frac{2}{3}$

B $\frac{2}{9}$

C $\frac{1}{6}$

D $\frac{1}{18}$

Mark Scheme & Explanation:

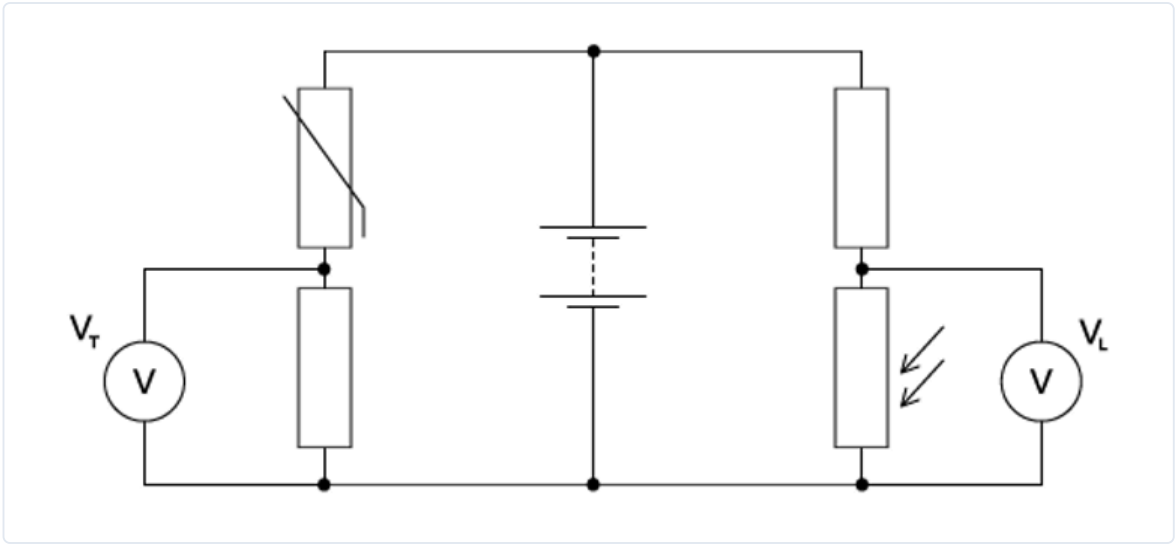
*The correct answer is **D** because:*

- Ohm's law states $V = IR$ or $I = \frac{V}{R}$
- From Kirchhoff's Second Law the p. d. is the same across P and Q
- So, ratio $\frac{I_P}{I_Q} = \frac{R_Q}{R_P}$
- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Cross-sectional area is equal to $A = \pi r^2 = \pi\frac{d^2}{2} = \frac{\pi d^2}{4}$
    - So, $R = \frac{\rho L}{A} = \rho L\frac{4}{\pi d^2}$

- For wire Q, the length is *L*, and the diameter is 3*d*
- So, the resistance of wire Q is equal to
  - $R_Q = \rho L \frac{4}{\pi 3d^2} = \frac{4}{9}\frac{\rho L}{\pi d^2}$
- For wire P, the length is 2*L*, and the diameter is *d*
- So, the resistance of wire P is equal to
  - $R_P = \rho 2L \frac{4}{\pi d^2} = 8\frac{\rho L}{\pi d^2}$
- Ratio $\frac{I_P}{I_Q} = \frac{R_Q}{R_P} = \frac{4}{9}\frac{\rho L}{\pi d^2} \div 8\frac{\rho L}{\pi d^2} = \frac{4}{9} \times \frac{1}{8} = \frac{1}{18}$

Question 32 Medium [1 mark]

In the circuit below, the reading V_T on the voltmeter changes from high to low as the temperature of the thermistor changes. The reading V_L on the voltmeter changes from high to low as the level of light on the light-dependent resistor (LDR) changes.



The readings V_T and V_L are both high.

What are the conditions of temperature and light level ?

	Temperature	Light level
A	low	low
B	low	high
C	high	low
D	high	high

A

B

C

D

Mark Scheme & Explanation:

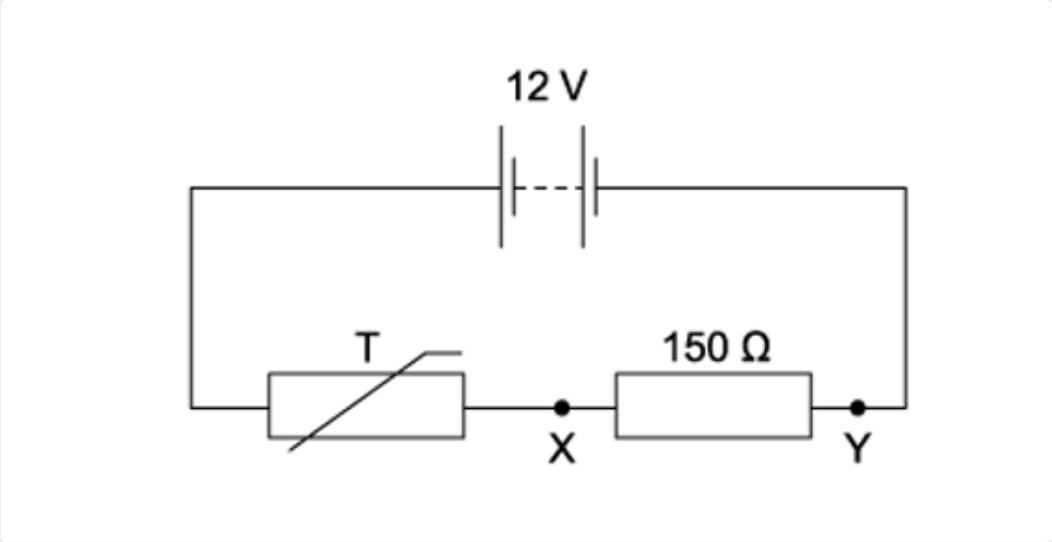
The correct answer is ***C*** *because:*

- Ohm's law states $V = IR$

- Therefore, for the voltmeter to read a high value, the resistance of the component to which it is connected must be high
- A thermistor is a type of resistor whose resistance varies significantly with temperature; generally, the resistance decreases with an increase in temperature
    - So, for the reading on V_T to be high, the resistance of the thermistor should be low meaning **the temperature must be high**
- A light-dependent resistor (LDR) is a light-controlled variable resistor. The resistance of an LDR decreases with increasing incident light intensity
    - For the reading on V_L to be high, the resistance of LDR should be high meaning **the light level must be low**

Question 33 Medium [1 mark]

In a fire alarm system, a thermistor T has a resistance of 2000 Ω at room temperature. Its resistance decreases as the temperature increases. The alarm is triggered when the potential difference between X and Y reaches 4.5 V.



What is the resistance of the thermistor when the alarm is triggered ?

A 90 Ω

B 150 Ω

C 250 Ω

D 1300 Ω

Mark Scheme & Explanation:

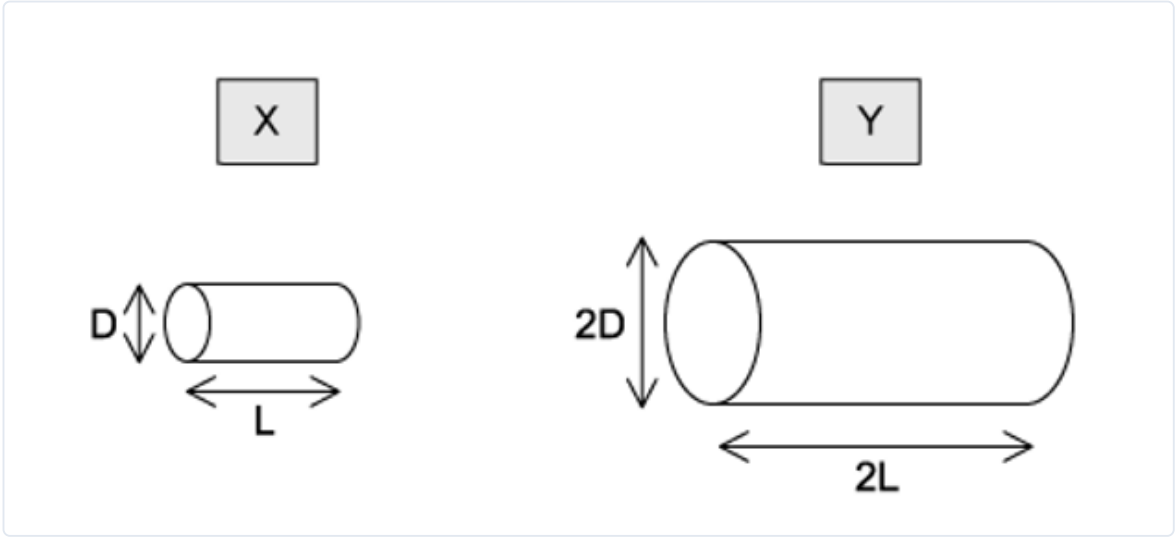
*The correct answer is **C** because:*

- Ohm's law states $V = IR$
- From Kirchhoff's Second Law, the e. m .f. is shared between the thermistor and the 150 Ω resistor
  - So, when the p. d. across the 150 Ω resistor is 4.5 V, the p. d. across the thermistor is 12 - 4.5 = 7.5 V
- From Kirchhoff's First Law, the current through both components is the same
  - So, the current in the circuit is $I = \frac{V}{R} = \frac{4.5}{150} = 0.03 \text{ A}$
- Therefore, the resistance of the thermistor is $R = \frac{V}{I} = \frac{7.5}{0.03} = 250\ \Omega$

Question 34 Medium [1 mark]

Two electrically-conducting cylinders X and Y are made from the same material.

Their dimensions are as shown.



The resistance between the ends of each cylinder is measured.

What is the ratio $\frac{resistance\ \ of\ \ X}{resistance\ \ of\ \ Y}$?

A $\frac{2}{1}$

B $\frac{1}{1}$

C $\frac{1}{2}$

D $\frac{1}{4}$

Mark Scheme & Explanation:

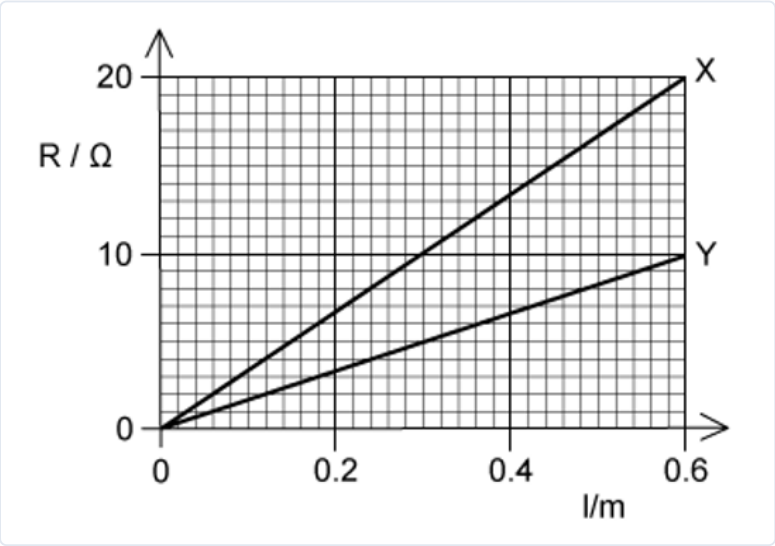
*The correct answer is **A** because:*

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Cross-sectional area is equal to $A = \pi r^2 = \pi \frac{d}{2}^2 = \frac{\pi d^2}{4}$
  - So, $R = \frac{\rho L}{A} = \rho L \frac{4}{\pi d^2}$
- For wire X, the length of the wire is *L*, and the diameter is *D*
  - $R_X = \rho L \frac{4}{\pi D^2} = 4\frac{\rho L}{\pi D^2}$

- For wire Y, the length of the wire is 2*L*, and the diameter is 2*D*
  - $R_Y = \rho 2L \frac{4}{\pi 2D^2} = 2\frac{\rho L}{\pi D^2}$
- So, ratio $\frac{R_X}{R_Y} = 4\frac{\rho L}{\pi D^2} \div 2\frac{\rho L}{\pi D^2} = \frac{2}{1}$

Question 35 Medium [1 mark]

The graph shows the variation with length *l* of resistance *R* for two wires X and Y made from the same material.



What does the graph show?

A cross-sectional area of X = 2 × cross - sectional area of Y

B resistivity of X = 2 × resistivity of Y

C when equal lengths of X and Y are connected in series to a battery, power in X = 2 × power in Y

D when equal lengths of X and Y are connected in parallel to a battery, current in X = 2 × current in Y

Mark Scheme & Explanation:

The correct answer is ***C*** *because:*

- Electrical power is given by $P = VI$
- Using Ohm's Law we can combine these to get
  - $P = IRI = I^2R$
- We can see from the graph, when *l* is 0.6 m:
  - The resistance of X is 20 Ω
  - The resistance of Y is 10 Ω
- Therefore, the resistance of X is double that of Y
- Since $P = I^2R$, power and resistance are directly proportional to each other

- The ratio of power in X to power in Y is $\frac{P_X}{P_Y} = \frac{20\ \ I^2}{10\ \ I^2} = 2$

- Therefore, when equal lengths of X and Y are connected in series to a battery, the power in X is double the power in Y

A is incorrect because from $R = \frac{\rho}{A}l$, we can see the gradient of the line is $\frac{\rho}{A}$, since the resistivity is the same for both wires, the cross-sectional area in Y will be double that of X

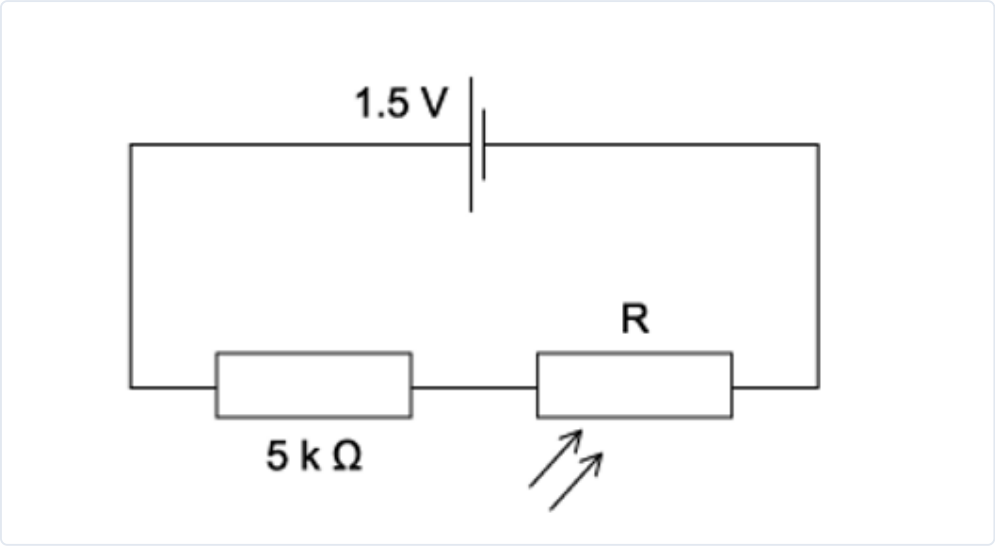
B is incorrect because the wires are made from the same material; hence, they both have the same resistivity

D is incorrect because from $R = \frac{V}{I}$, we can see resistance and current are inversely proportional, so the current in Y will be double the current in X

Exam Tip: Sometimes, in physics questions, you can use common sense to work out whether or not your answer makes logical sense. In this question, for example, as the resistance in wire X is double that of wire Y, it makes sense that the cross-sectional area, and hence, the current will be lower in wire X. Think about water flowing through a pipe - a smaller cross-sectional area means a thinner pipe, and hence water will travel at a slower rate than in a wider pipe (one with a larger cross-sectional area)

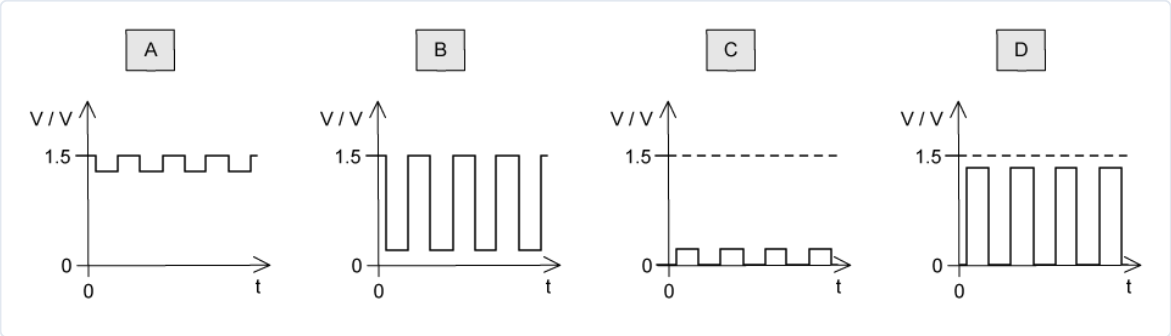
Question 36 Medium [1 mark]

A light-dependent resistor *R* has a resistance of about 1 MΩ in the dark and about 1 kΩ when illuminated. It is connected in series with a 5 kΩ resistor to a 1.5 V cell of negligible internal resistance.



The light-dependent resistor is illuminated (in an otherwise dark room) by a flashing light.

Which graph best shows the variation with time *t* of potential difference *V* across *R* ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The LDR and the 5 kΩ resistor form a potential divider
- The potential divider equation is given by $V_{emf} \times \frac{R_1}{R_1 + R_2}$, where:
  - V_{emf} is the e. m .f. of the supply = 1.5 V

- R_1 is the resistance across the LDR = R
- R_2 is the resistance of the fixed resistor = 5 kΩ

- So, the p. d. across the LDR is equal to $\frac{1.5 \quad R}{R + 5000}$
- When R is 1000 kΩ, the p. d. is very close to 1.5 V:
  - $V = 1.5 \times \frac{1 \times 10^6}{1 \times 10^6 + 5000} = 1.492 \approx 1.5 \quad \text{V}$
- When R is 1 kΩ, then the p. d. is about $\frac{1.5}{6}$:
  - $V = 1.5 \times \frac{1000}{1000 + 5000} = 0.25 \quad \text{V}$
- So, the graph must show a peak value of very close to 1.5 and a minimum value that is more than zero

A is incorrect because this graph shows the minimum to be only slightly less than 1.5 V

C is incorrect because this graph shows the minimum to be zero

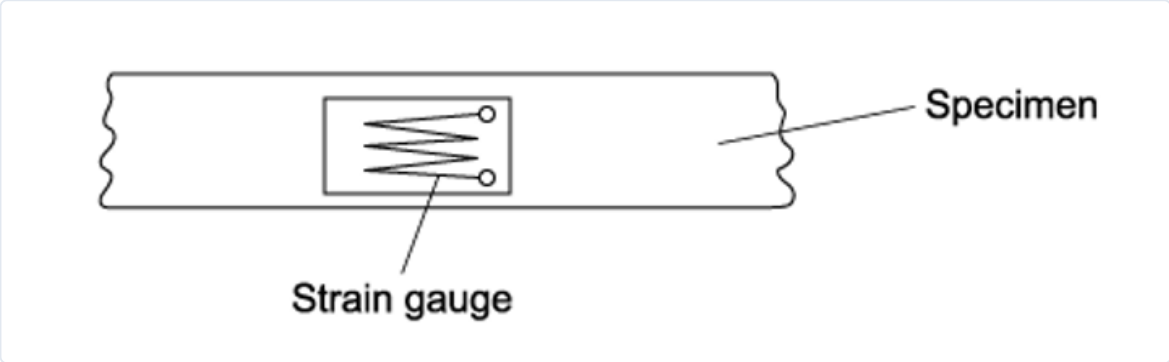
D is incorrect because the maximum is far below 1.5 V

Question 37 Medium [1 mark]

Tensile strain may be measured by the change in electrical resistance of a strain gauge.

A strain gauge consists of folded fine metal wire mounted on a flexible insulating backing sheet.

The strain gauge is firmly attached to the specimen, so that the strain in the metal wire is always identical to that in the specimen.



When the strain in the specimen is increased, what happens to the resistance of the wire?

A it decreases, because the length decreases and the cross-sectional area increases

B it decreases, because the length increases and the cross-sectional area decreases

C it increases, because the length decreases and the cross-sectional area increases

D it increases, because the length increases and the cross-sectional area decreases

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- As the strain in the specimen is increased, the wire is extended meaning its length increases
- Resistance and wire length are directly proportional, so an increase in length results in an increase in resistance
- At the same time, the strain causes the cross-sectional area of the wire to become smaller (as the overall volume does not change)
- Resistance and cross-sectional area are inversely proportional, so, a decrease in cross-sectional area also results in an increase in the resistance

- So, as the strain in the specimen increases, the resistance of the wire increases, because the length increases and the cross-sectional area decreases

Question 38 Medium [1 mark]

The resistance of a thermistor depends on its temperature, and the resistance of a light-dependent resistor (LDR) depends on the illumination.

Under which conditions will the resistance of both a thermistor and an LDR be highest ?

	Thermistor	LDR
A	highest temperature	highest illumination
B	highest temperature	lowest illumination
C	lowest temperature	highest illumination
D	lowest temperature	lowest illumination

A

B

C

D

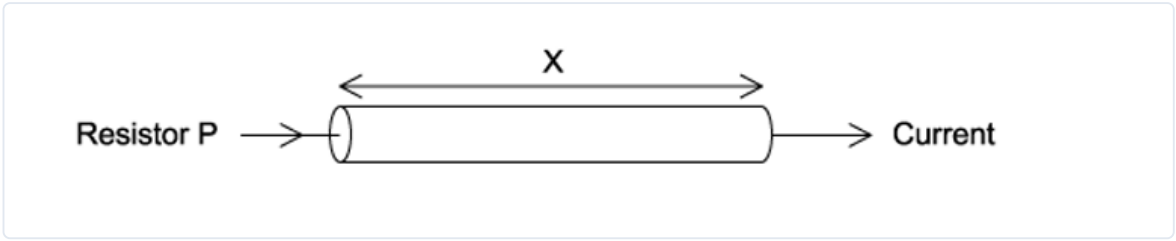
Mark Scheme & Explanation:

*The correct answer is **D** because:*

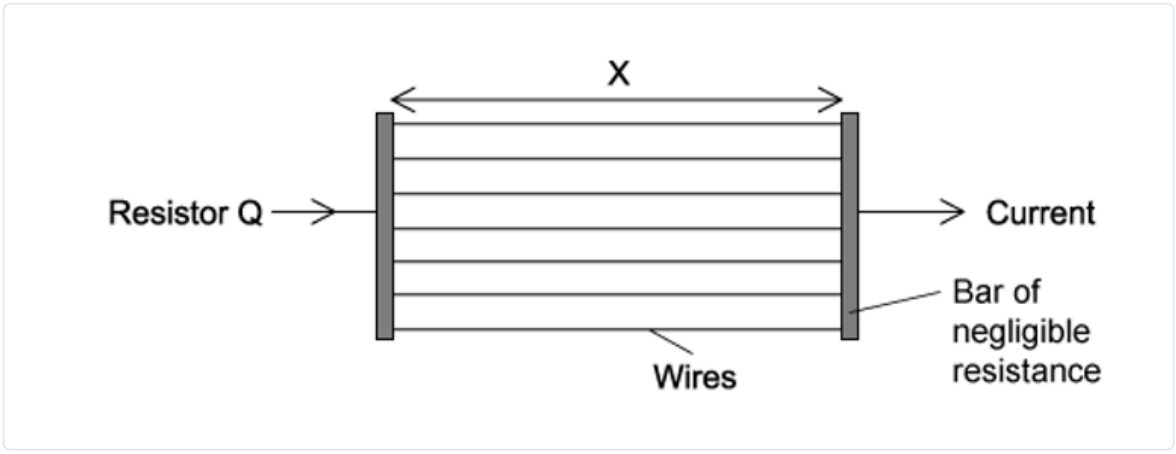
- A thermistor is a type of resistor whose resistance varies significantly with temperature; generally, the resistance decreases with an increase in temperature
  - So, for the resistance of the thermistor to be high, the temperature must be low
- A light-dependent resistor (LDR) is a light-controlled variable resistor. The resistance of an LDR decreases with increasing incident light intensity
  - For the resistance of an LDR to be high, the light level must be low

Question 39 Medium [1 mark]

A researcher has two pieces of copper of the same volume. All of the first piece is made into a cylindrical resistor P of length *x*.



All of the second piece is made into uniform wires each of the same length *x* which he connects between two bars of negligible resistance to form a resistor Q.



How do the electrical resistances of P and Q compare ?

A P has a larger resistance than Q

B Q has a larger resistance than P

C P and Q have equal resistance

D Q may have a larger or smaller resistance than P, depending on the number of wires made

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Let's compare each of these quantities
    - The wires are all made of the same material, so the resistivity ρ is the same for both

    - Both wires have the same length *x*
    - The only difference between the two resistors is the number of wires in each configuration
- Both resistors contain the same amount (volume) of material, so the cross-sectional area *A* of resistor P will equal to collective cross-sectional areas of all the wires in resistor Q
- Therefore, since none of the quantities in $R = \frac{\rho L}{A}$ are different, P and Q both have the same resistance

Question 40 Medium [1 mark]

Two wires have the same length and the same resistance. Wire X is made of a metal of resistivity 1.7×10^{-8} Ωm, and wire Y is made of a metal of resistivity 5.6×10^{-8} Ωm.

The diameter of wire X is 0.315 mm.

What is the diameter of wire Y?

A 0.17 mm

B 0.33 mm

C 0.57 mm

D 1.0 mm

Mark Scheme & Explanation:

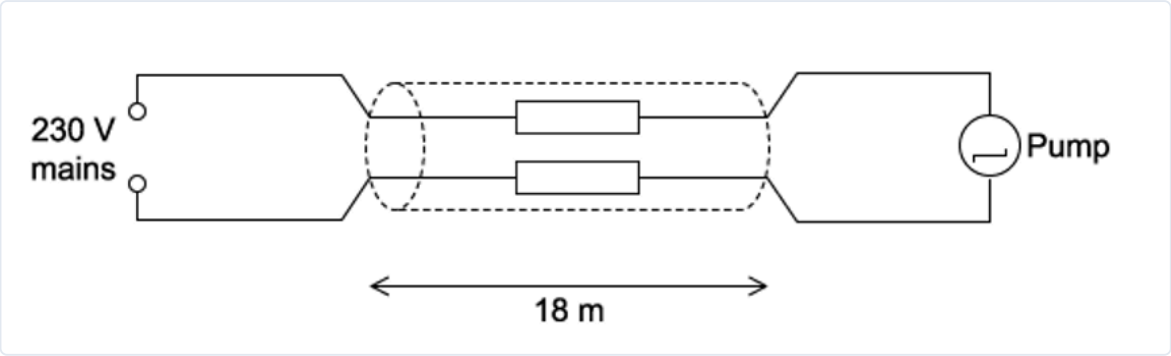
*The correct answer is **C** because:*

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Cross-sectional area is equal to $A = \pi r^2 = \pi \frac{d}{2}^2 = \frac{\pi d^2}{4}$
  - So, $R = \frac{\rho L}{A} = \rho L \frac{4}{\pi d^2}$
- The two wires have the same resistance and same length
  - Resistance in wire X $= \frac{41.7 \times 10^{-8} L}{\pi 0.315^2}$
  - Resistance in wire Y $= \frac{45.6 \times 10^{-8} L}{\pi d^2}$
- Resistance in wire X = Resistance in wire Y
  - $\frac{41.7 \times 10^{-8} L}{\pi 0.315^2} = \frac{45.6 \times 10^{-8} L}{\pi d^2}$
- Cancelling the common terms $\frac{4L}{\pi}$, we have
  - $\frac{1.7 \times 10^{-8}}{0.315^2} = \frac{5.6 \times 10^{-8}}{d^2}$
- Rearranging for d^2
  - $d^2 = \frac{5.6 \times 10^{-8} 0.315^2}{1.7 \times 10^{-8}} = 0.327$
- Therefore, the diameter of wire Y is

- $d = \sqrt{0.327} = 0.57 \quad \text{mm}$

Question 41 Hard [1 mark]

The diagram shows an electric pump for a garden fountain connected by an 18 m cable to a 230 V mains electrical supply.



The performance of the pump is acceptable if the potential difference (p.d.) across it is at least 218 V. The current through it is then 0.83 A.

What is the maximum resistance per meter of each of the two wires in the cable if the pump is to perform acceptably?

A 0.40 Ω m^{-1}

B 0.80 Ω m^{-1}

C 1.3 Ω m^{-1}

D 1.4 Ω m^{-1}

Mark Scheme & Explanation:

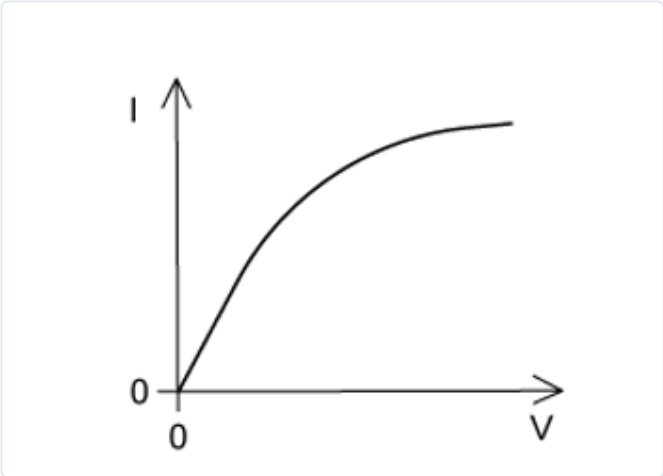
*The correct answer is **A** because:*

- Ohm's Law states $V = IR$
- For the pump to perform acceptably, we are told the potential difference across it should be at least 218 V
  - So, the potential difference across the cables is 230 -218 = 12 V
- The current through the electric pump is 0.83 A
- The same current flows through the cable since it is in series with the pump. So, the cable has a total resistance of $R = \frac{230 - 218}{0.83} = 14.5\Omega$
- From the diagram, it can be seen that the cable is made up of two wires
  - So, the total length of wire in the cable is 2 × 18 = 36 m

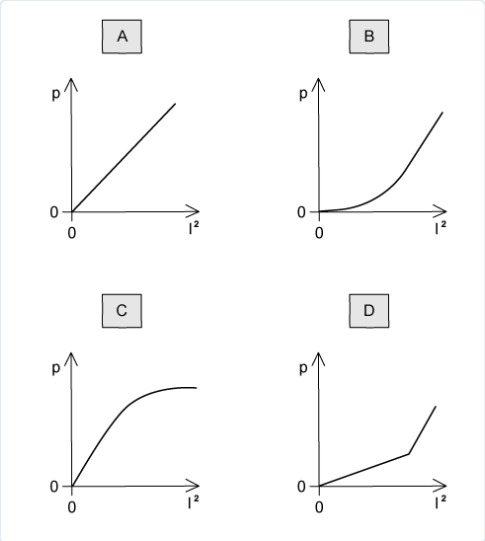
- The wires are not connected to each other - so, they are not in parallel. Instead, one wire connects one terminal of the mains to one terminal of the pump, and the other wire connected the other terminal of the mains to the other terminal of the pump
- This total length of wire corresponds to a total resistance of 14.5 Ω
    - Therefore, the resistance per meter is equal to $= \frac{14.5}{36} = 0.4 \Omega \text{ m}^{-1}$

Question 42 Hard [1 mark]

The graph below shows the variation of the current *I* through a lamp with potential difference *V* across it.



Which of the following graphs best represents the variation of power *P* dissipated in the same lamp with I^2?



A

B

C

D

Mark Scheme & Explanation:

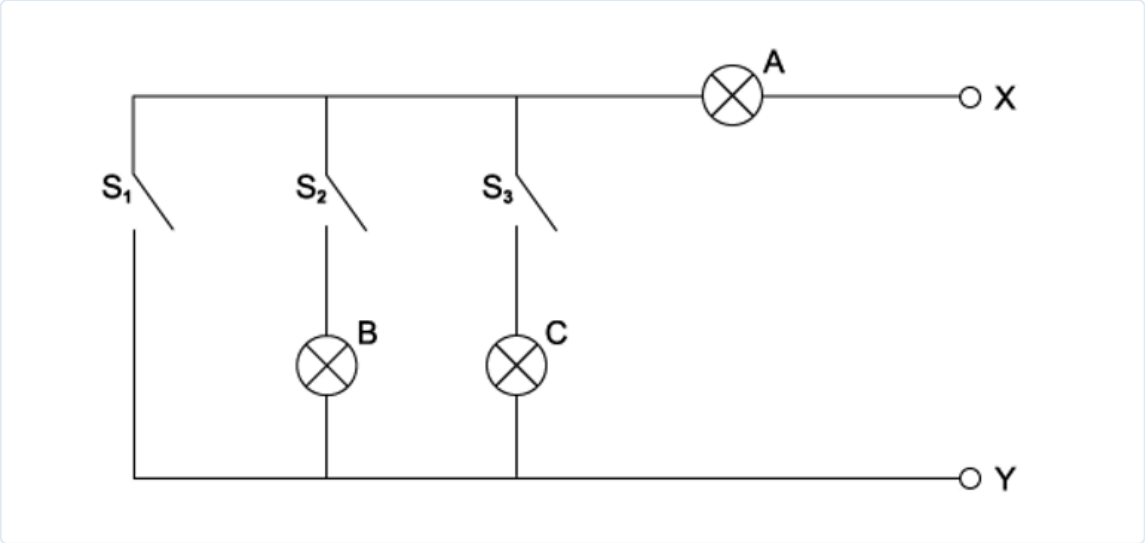
*The correct answer is **B** because:*

- Ohm's Law states that $V = IR$, or $R = \frac{V}{I}$
- This means the slope of the *I* -*V* graph is equal to $\frac{1}{R}$

- Therefore, as current *I* increases, *R* increases gradually as the slope decreases
- Electrical power is equal to $P = VI = IRI = I^2R$, or $R = \frac{P}{I^2}$
- The slope of a $P – I^2$ graph is equal to *R*
- Therefore, from the graph of $P – I^2$, as *R* increases the slope of $P – I^2$ increases as *I* increases
- Hence, the graph of $P – I^2$ will be the inverse of the *I* - *V* graph

Question 43 Hard [1 mark]

Circuit contains three identical lamps **A**, **B** and **C**. Circuit also contains three switches, S_1, S_2 and S_3, as shown.



One of the lamps is faulty. In order to detect the fault, an ohm-meter is connected between terminals X and Y.

When measuring resistance, the ohm-meter causes negligible current in the circuit. Readings of the ohm-meter for different switch positions are shown in the table.

Switch			Meter reading / Ω
S_1	S_2	S_3	
open	open	open	∞
closed	open	open	15
open	closed	open	30
open	closed	closed	15

Which lamps are faulty?

A A & B

B B only

C B & C

D C only

Mark Scheme & Explanation:

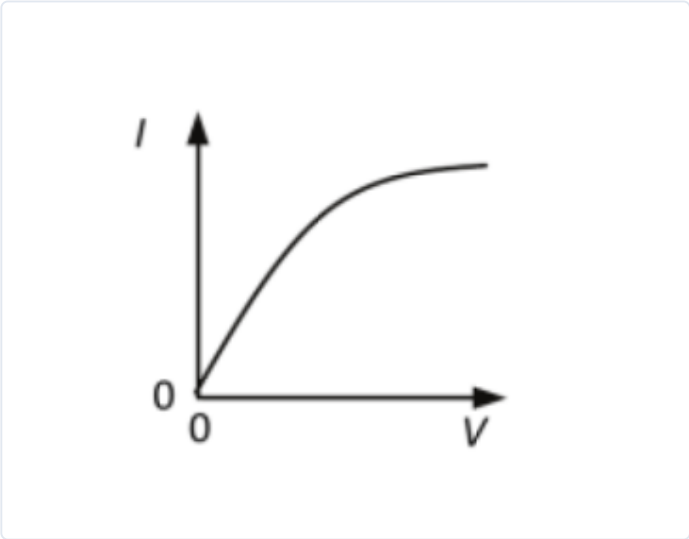
*The correct answer is **D** because:*

- In the first test, when all the switches are open, the meter reads infinity
  - This is obvious since the circuit is incomplete and no current flows
- In the second test, where only S_1 is closed, we can see that lamp A is not faulty
  - This is because the ohm-meter reads 15Ω, proving that a current is passing through component A
- In the third test where S_1 is open, S_2 is closed and S_3 is open, the meter reads a resistance of 30Ω, so, this test proves that lamp B is also not faulty
  - This is because components A and B are now connected in series, giving a total resistance of 15 + 15 = 30Ω
- In the last test, where only S_1 is open, the resistance reading is 15Ω
- This proves that lamp C is shorted, meaning the current passes through as if there was no lamp
  - In this case, it's as if a wire of negligible resistance is connected in parallel with lamp B, and since Ohm's Law , $I \ \alpha \ \frac{1}{R}$ the current is larger when the resistance is smaller
  - So, all current would flow in that wire instead of through lamp B (since lamp B has some resistance)
  - Now, the only resistance encountered by the current is due to lamp A only, so, that's why the reading on the ohm-meter is 15Ω

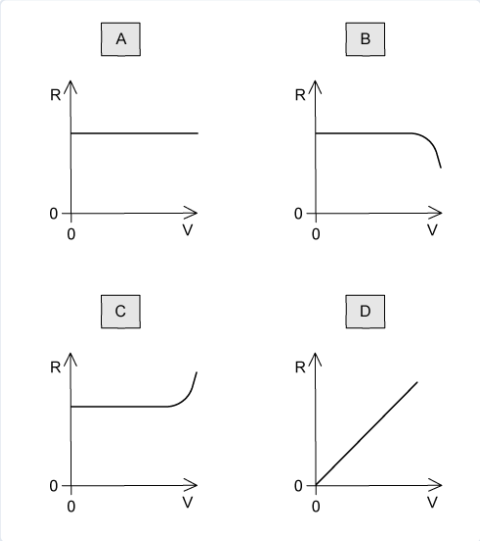
Exam Tip: Remember current flow in a circuit will always take the route with the least resistance. If S_1 and S_2 were closed, and S_3 was opened, the ohm-meter would again read 15Ω since all the current would pass through S_1 (which has no resistance) instead of passing through lamp B (which has some resistance) and therefore, the current would then flow through lamp A (which is the only lamp that causes a resistance in this case)

Question 44 Hard [1 mark]

The current flowing through a component varies with the potential difference V across it as shown.



Which graph best represents how the resistance R varies with V ?



A

B

C

D

Mark Scheme & Explanation:

The correct answer is ***C*** *because:*

- Ohm's Law states that $V = IR$, or $R = \frac{V}{I}$

- This means the slope of the *I* - *V* graph is equal to $\frac{1}{R}$
- The *I*-*V* graph consists of two sections
    - The first section at lower voltages has constant resistance because it is a linear *I* -*V* relationship
    - The second half at higher voltages has increasing resistance because the gradient is decreasing
- Therefore, in an R–V graph, initially the line would be horizontal as $\frac{V}{I}$ is constant, then, the line will begin to curve upwards as *R* increases
    - Option C corresponds best to this description

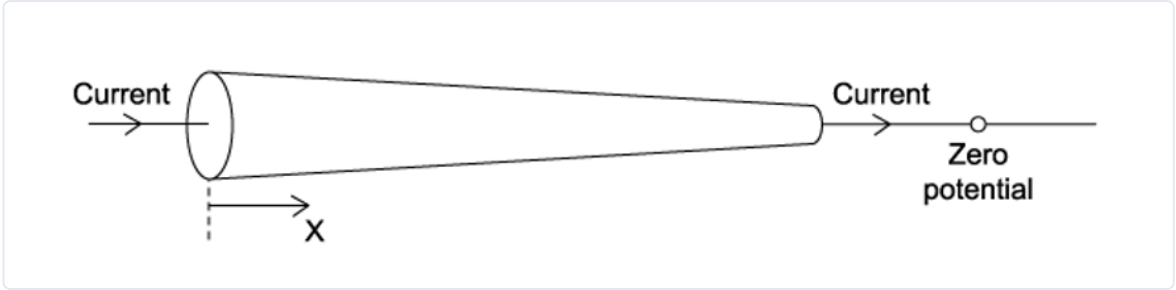
A is incorrect because the resistance does not stay constant, or the *I* - *V* graph would be completely straight

B is incorrect because even though the first section of this graph is correct, the second half shows the resistance is decreasing in the second half, which is incorrect

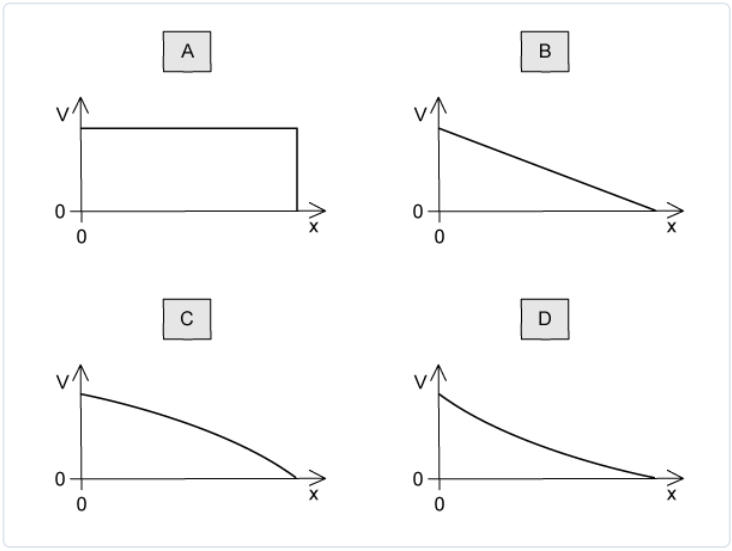
D is incorrect because the first half of the *I* - *V* graph shows the component has an initial resistance which is constant, it does not begin at zero and it does not increase linearly

Question 45 Hard [1 mark]

The circular cross-sectional area of a uniform metal wire varies along its length. There is a current in the wire. The narrow end of the wire is at a reference potential of zero.



Which graph best represents the variation with distance *x* along the wire of the potential difference *V* relative to the reference zero?



A

B

C

D

Mark Scheme & Explanation:

The correct answer is ***C*** *because:*

- Ohm's Law states $V = IR$
    - In this case, the current is constant along the wire, only the resistance changes and this causes a proportional change in the potential difference
- Resistance of wire is equal to $R = \frac{\rho L}{A}$

- When the length of wire *L* is at a maximum distance from the zero potential reference, resistance *R* will also be at a maximum and hence, potential difference *V* too (since $R \alpha L \alpha V$)
    - So, as distance *x* increases, the length *L* decreases, and so, *R* and *V* also decrease
- Now, the diameter of the wire is falling linearly as *x* increases, but the area of cross-section A is not falling linearly as $A = \pi(\frac{d}{2})^2$ where *d* is the diameter
- Since resistance *R* is inversely proportional to cross-sectional area *A* $R \alpha \frac{1}{A}$ it is also inversely proportional to d^2 (when *d* is small, *R* is large)
- Therefore, there is a greater percentage fall (in resistance, and hence, potential difference) per unit length at the narrow end than at the wide end, meaning the decrease is not linear, but depends on d^2
- $\frac{V}{x}$This means that potential difference per unit length (this is represented by the gradient of the graphs shown which equal to) is less at the wide end than at the narrow end (the gradient at the highest value of *x* should be greater than the gradient at *x* = 0 and the gradient keeps on increasing as *x* increases)

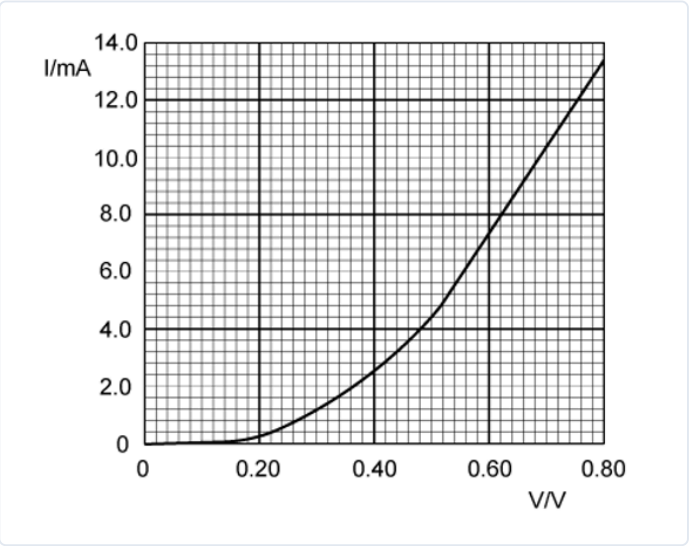
A is incorrect because *V* decreases with distance, whereas this graph shows it stays constant

B is incorrect because *V* depends on d^2 and therefore does not decrease linearly

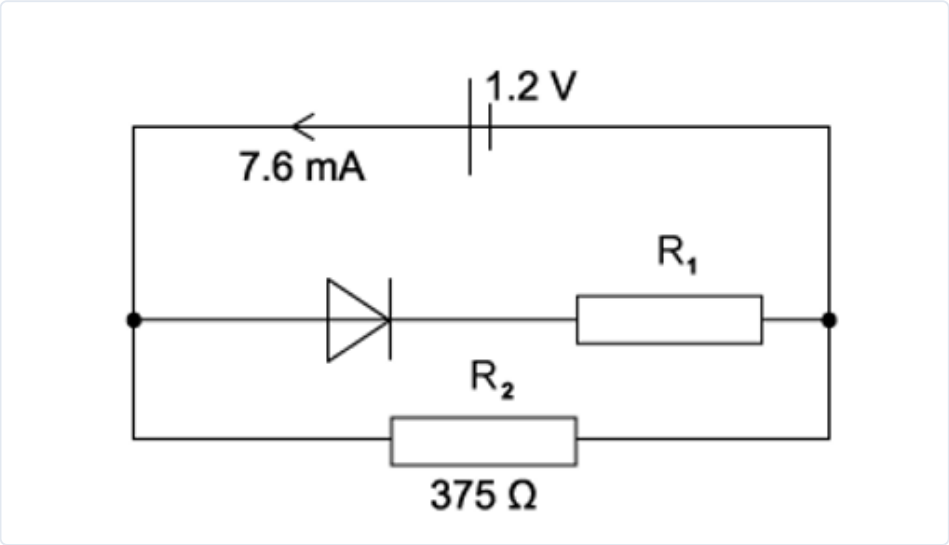
D is incorrect because the gradient is highest at *x* = 0 and gradually increases

Question 46 Hard [1 mark]

The graph shows the *I* -*V* characteristics of a semiconductor diode



A cell of e. m .f. 1.2 V and negligible internal resistance is connected in series to a semiconductor diode and a resistor R_1, as shown below



A resistor R_2 of resistance 375 Ω is connected across the cell. The diode has the characteristic shown in the graph. The current supplied by the cell is 7.6 mA.

What is the resistance of R_1?

A 82 Ω

B 114 Ω

C 160 Ω

D 260 Ω

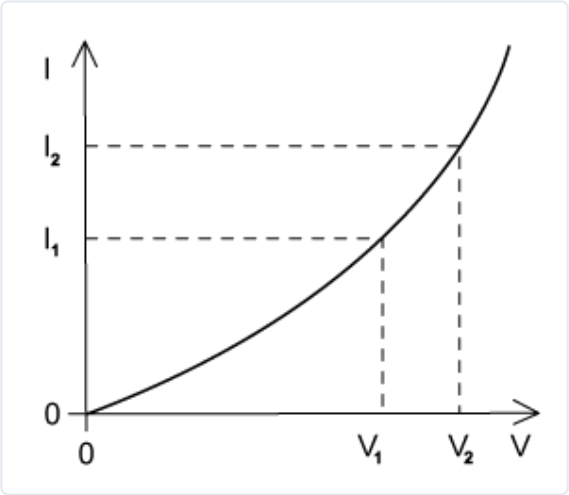
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The p.d. across each loop is equal to the e. m. f. in the circuit
  - Ohm's Law is $I = \frac{V}{R}$
- So, the current across the 375 Ω resistor is $\frac{1.2}{375} = 3.2 \times 10^{-3} \quad A = 3.2 \quad \text{mA}$
  - This means the current through the semiconductor diode and R_1 is 7.6 – 3.2 = 4.4 mA
- To find the p. d. across the semiconductor diode we need to use the graph
  - For a current of 4.4 mA, the p. d. from graph is 0.5 V
- So, p. d. across R_1 = 1.2 – 0.5 = 0.7 V
  - Resistance is equal to $R = \frac{V}{I} = \frac{0.7}{4.4 \times 10^{-3}} = 159\ \Omega$
- Therefore, the resistance across R_1 is 160 Ω

Question 47 Hard [1 mark]

The graph shows how the current *I* varies with potential difference *V* for an electrical component.



Which row of the table gives the resistance of the component at V_2 and describes how the resistance changes from V_1 to V_2?

	Resistance at V_2	Change in resistance from V_1 to V_2
A	$\frac{V_2 - V_1}{I_2 - I_1}$	increases
B	$\frac{V_2 - V_1}{I_2 - I_1}$	decreases
C	$\frac{V_2}{I_2}$	increases
D	$\frac{V_2}{I_2}$	decreases

A

B

C

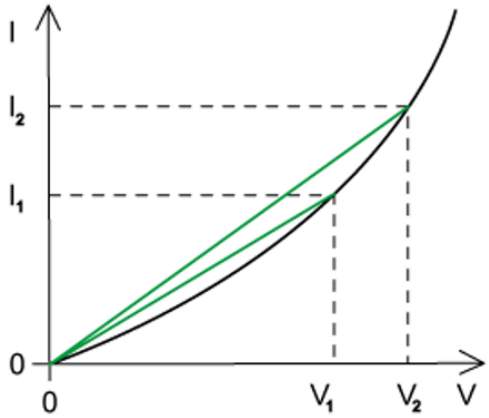
D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

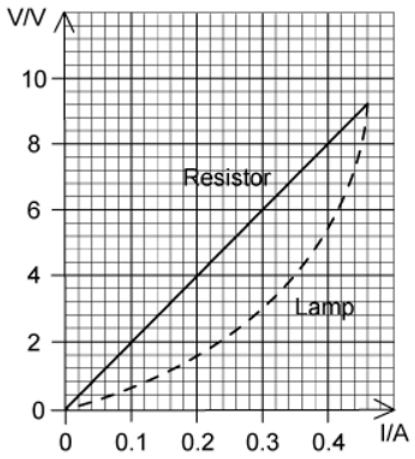
- Ohm's Law states that $V = IR$, or $R = \frac{V}{I}$

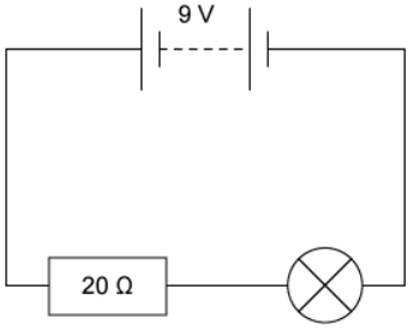
- The resistance at V_2 is simply found by reading the values of current and p. d. from the axes, **not** from taking the tangent between the two points
- This is because resistance is defined by $\frac{V}{I}$ **not** $\frac{\Delta V}{\Delta I}$
    - So, the resistance at V_2 is equal to $\frac{V_2}{I_2}$
- To see how the resistance changes from V_1 to V_2 we have to consider the tangents from the **origin** to the two points (not the tangents on the curve)
- From Ohm's Law, we know the slope of the I - V graph is equal to $\frac{1}{R}$
    - So, the steeper the gradient, the smaller the value of resistance
- The gradient gets steeper from V_1 to V_2, meaning the resistance is decreasing



Question 48 Hard [1 mark]

The graph shows the relationship between potential difference *V* and current *I* for a fixed 20 Ω resistor and a filament lamp.





The resistor and lamp are placed in series with a 9 V battery of negligible internal resistance. What is the current in the circuit ?

A 0.1 A

B 0.2 A

C 0.3 A

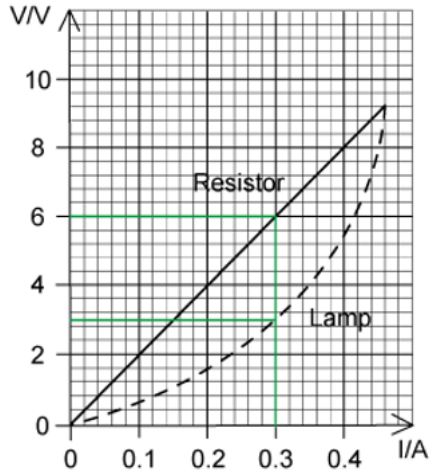
D 0.4 A

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- In this question, we must look at the values of potential difference (p.d.) of each component at each current to determine which two values of p. d. add up to make 9 V
- Consider option C, at 0.3 A:
  - The p. d. of the resistor is 6 V
  - The p .d. of the lamp is 3 V
- In a series circuit, the e. m. f from the battery is split between the components, so we just add them together
  - So, the total p. d. of the components is 6 + 3 = 9 V

- This is the e. m .f of the battery, so the current is 0.3 A



A is incorrect because p .d. of the resistor = 2 V, p. d. of the lamp = 0.6 V

Total p d. = 2.6 V

B is incorrect because p .d. of the resistor = 4 V, p. d. of the lamp = 1.6 V

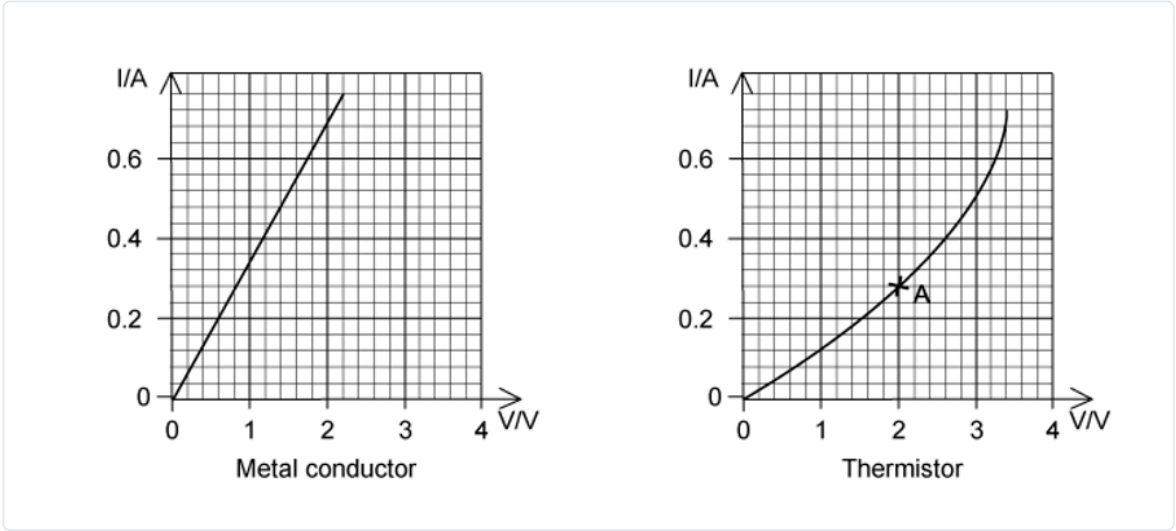
Total p .d. = 5.6 V

D is incorrect because p. d. of the resistor = 8 V, p d. of the lamp = 5.4 V

Total p. d. = 13.4 V

Question 49 Hard [1 mark]

The graphs show the *I* -*V* characteristics for a metal conductor and a thermistor



What is the ratio $\frac{resistance\ \ of\ \ thermistor\ \ at\ \ point\ \ A}{resistance\ \ of\ \ metal\ \ conductor}$?

A $\frac{2}{5}$

B $\frac{1}{2}$

C $\frac{2}{1}$

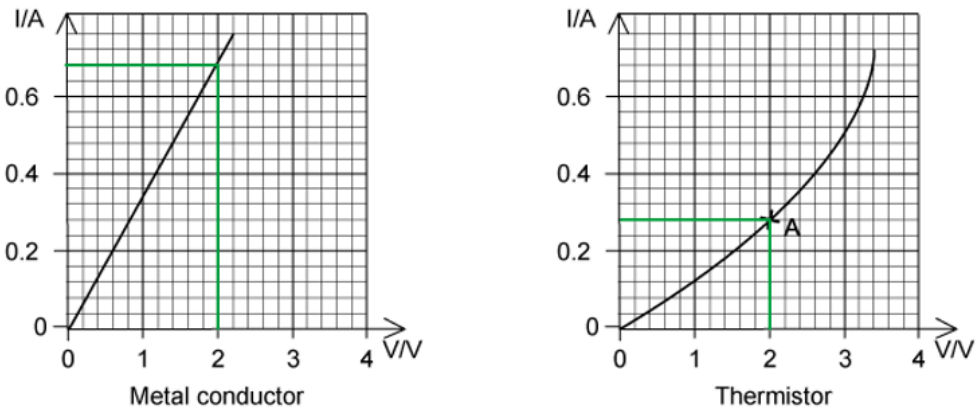
D $\frac{5}{2}$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Ohm's Law states that $V = IR$, or $R = \frac{V}{I}$
- For the metal conductor:
  - At *V* = 2 V, the current is *I* = 0.68 A
- So, the resistance of the metal conductor is:
  - $R_M = \frac{V}{I} = \frac{2}{0.68} = \frac{50}{17} = 3\Omega$
- For the thermistor, at point A:
  - At *V* = 2 V, the current is *I* = 0.27 A

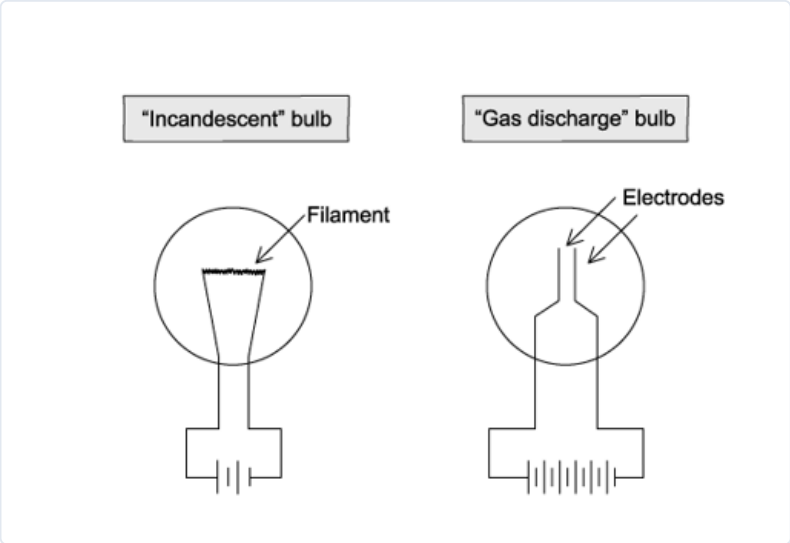
- The resistance at point A is simply found by reading the values of current and p. d. from the axes, **not** from taking the tangent of this point on the curve, so the resistance of the thermistor is:
  - $R_T = \frac{V}{I} = \frac{2}{0.27} = \frac{200}{27} = 7.4\ \Omega$
- So, the ratio is given by :
  - $\frac{R_T}{R_M} = \frac{200}{27} \div \frac{50}{17} = \frac{5}{2}$



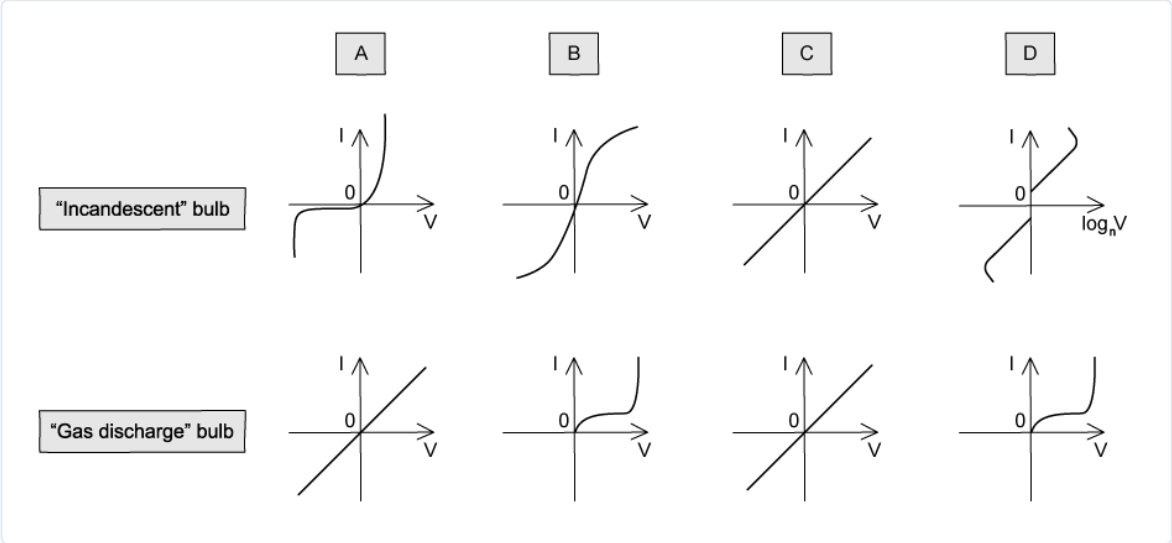
Question 50 Hard [1 mark]

One style of light bulb, very different from the "incandescent" design which works on the principle of a super-heated wire filament emitting light, is called a gas discharge tube.

In this design of light bulb, light is produced by the direct excitation of gas molecules as electric current passes between two electrodes.



What will the *I* - *V* graphs of these bulbs look like?



A

B

C

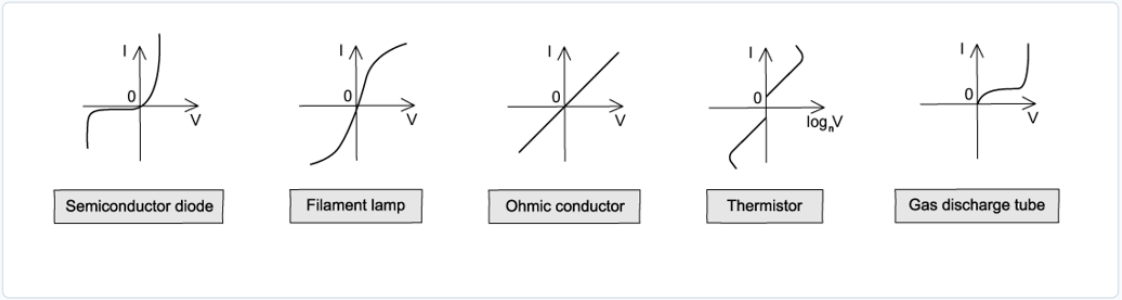
D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- An incandescent bulb is equivalent to a filament lamp, in which the resistance increases as V increases
- You are not expected to know what the graph of a gas discharge bulb looks like, but just to realize it is non-Ohmic
- A device is said to be non-Ohmic if the resistance across it is not constant, but changes in a way that depends on the voltage across it
- In this case, the I-V graph is not a straight line but has some curvy shape. Devices such as this do not have a constant value of resistance; the resistance is called dynamic resistance because it is constantly changing as V and I change

The *I-V* graphs are for the following components:

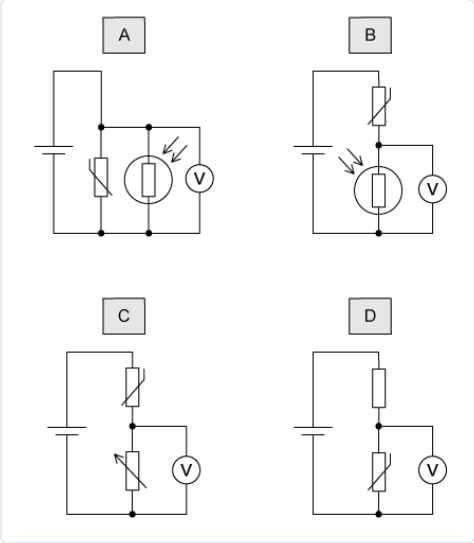


Exam Tip: Note the graph for the thermistor is plotted as *I* against log *V*, making it easier to plot large values of *V*. Don't let this confuse you, logs are just another way of writing exponential equations, one that allows you to separate the exponent on one side of the equation. For example, the equation $2^4 = 16$ can be rewritten as $\log_2(16) = 4$ and expressed as "log to the base 2 of 16 is 4". Rest assured; you will not be expected to deal with logarithmic scales in the AS course.

Question 51 Hard [1 mark]

A thermistor and another component are connected to a constant voltage supply. A voltmeter is connected across one of the components. The temperature of the thermistor is then reduced, but no other changes are made.

In which circuit will the voltmeter reading increase ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The resistance of a thermistor decreases as the temperature increases
- The temperature of the thermistor is reduced, and no other changes are made, so, its resistance increases
- From Ohm's law, $V = IR$, an increase in the resistance of the thermistor would cause the p.d. across it to increase
- Therefore, for the voltmeter reading to increase, the voltmeter should be connected in parallel to the thermistor so that it reads the p. d. across the thermistor

A is incorrect because the voltmeter reading remains constant since all three components are connected in parallel to each other

B is incorrect because the resistance of an LDR decreases as the light intensity increases, but in this case, the light intensity is unchanged, so its resistance is unchanged

C is incorrect because the voltmeter is across the variable resistor so the increase in resistance from the thermistor would cause a decrease in resistance across the variable resistor

Question 52 Hard [1 mark]

A copper wire is to be replaced by an aluminum alloy wire of the same length and resistance.

Copper has half the resistivity of the alloy.

What is the ratio $\frac{diameter\ \ of\ \ alloy\ \ wire}{diameter\ \ of\ \ copper\ \ wire}$?

A $\sqrt{2}$

B 2

C $\sqrt[2]{2}$

D 4

Mark Scheme & Explanation:

*The correct answer is **A** because:*

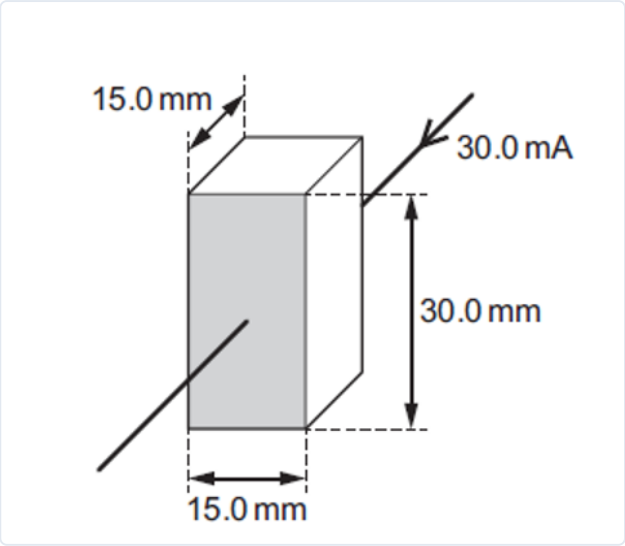
- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Cross-sectional area of a wire is equal to $A = \pi r^2 = \pi \frac{d^2}{2} = \frac{\pi d^2}{4}$
  - The resistance of the copper wire is equal to $R = \frac{\rho_c L}{A}$
  - So, the cross-sectional area is equal to $A = \frac{\pi d_c{}^2}{4}$
- We can combine the above equations to give
  - $R = \rho_c L \frac{4}{\pi d_c{}^2}$
- So, the diameter of the copper wire is equal to
  - $d_c{}^2 = \frac{4\rho_c L}{\pi R}$
  - $d_c = 2\sqrt{\frac{\rho_c L}{\pi R}}$
- We are told copper has half the resistivity of the alloy, so the resistivity of the alloy is equal to $2\rho_c$
  - The resistance of the aluminum wire is equal to $R = \frac{2\rho_c L}{A}$
  - So, the cross-sectional area is equal to $A = \frac{\pi d_A{}^2}{A}$
- We can combine these equations to give
  - $R = 2\rho_c L \frac{4}{\pi d_A{}^2}$
- So, the diameter of the aluminum wire is equal to

- $d_A{}^2 = \frac{8\rho_c L}{\pi R}$
- $d_A = 2\sqrt{2}\sqrt{\frac{\rho_c L}{\pi R}}$

- So, the ratio $\frac{d_A}{d_c} = \frac{2\sqrt{2}}{2} = \sqrt{2}$

Question 53 Hard [1 mark]

The current in a block of semiconductor is 30.0 mA when there is a potential difference (p.d.) of 10.0 V across it. The dimensions of the block and the direction of the current in it are as shown.



The electrical meters used are accurate to ± 0.1 mA and ± 0.1 V. The dimensions of the block are accurate to ± 0.2 mm.

What is the resistivity of the semiconductor?

A 10.0 ± 0.2 Ω m

B 10.0 ± 0.3 Ω m

C 10.0 ± 0.5 Ω m

D 10.0 ± 0.8 Ω m

Mark Scheme & Explanation:

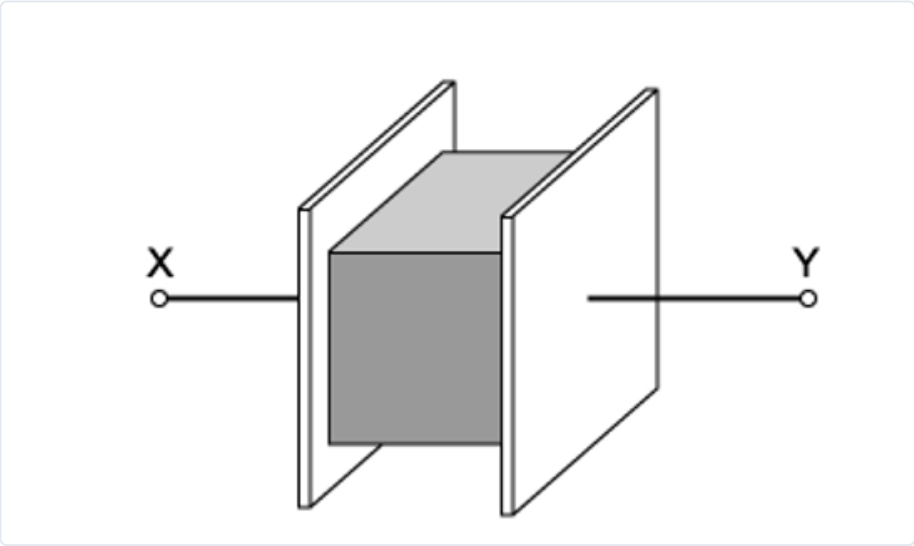
The correct answer is ***C*** *because:*

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- So, resistivity is equal to $\rho = \frac{AR}{L}$
- From Ohm's law, resistance is equal to $R = \frac{V}{I}$
- So, combining this with the equation for resistivity, we get
  - $\rho = \frac{VA}{IL}$

- The cross-sectional area of the block is equal to the area of the grey rectangle through which the current flows
    - $A = lw$, where l is the length and w is the width
- So, resistivity is equal to $\rho = \frac{Vlw}{IL}$
    - The p. d. is equal to V = 10.0 V
    - The current through the block is equal to I = 30 mA = 30 × 10^{-3} A
- The dimensions of the block are equal to
    - l = 15 × 10^{-3} m, w = 30 × 10^{-3} m, L = 15 × 10^{-3} m
- So, resistivity $\rho = \frac{Vlw}{IL} = \frac{10 \times 15 \times 10^{-3} \times 30 \times 10^{-3}}{30 \times 10^{-3} \times 15 \times 10^{-3}} = 10$ Ωm
- Next, we must consider the uncertainty in resistivity ρ
    - $\frac{\Delta\rho}{\rho} = \frac{\Delta V}{V} + \frac{\Delta l}{l} + \frac{\Delta w}{w} + \frac{\Delta I}{I} + \frac{\Delta L}{L}$
    - $\Delta\rho = \frac{\Delta v}{V} + \frac{\Delta l}{l} + \frac{\Delta w}{w} + \frac{\Delta I}{I} + \frac{\Delta L}{L} \times \rho$
    - $\Delta\rho = \frac{0.1}{10} + \frac{0.2}{15} + \frac{0.2}{30} + \frac{0.1}{30} + \frac{0.2}{15} \times 10$
    - $\Delta\rho$ = 0.467 = 0.5 Ωm
- Therefore, resistivity is equal to ρ = 10.0 ± 0.5 Ωm

Question 54 Hard [1 mark]

The resistance of a metal cube is measured by placing it between two parallel plates, as shown.



The cube has volume *V* and is made of a material with resistivity *ρ*. The connections to the cube have negligible resistance.

Which expression gives the electrical resistance of the metal cube between X and Y?

A $\rho V^{\frac{1}{3}}$

B $\rho V^{\frac{2}{3}}$

C $\frac{\rho}{V^{\frac{1}{3}}}$

D $\frac{\rho}{V^{\frac{2}{3}}}$

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Where *L* is the length of the wire and *A* is the cross-sectional area of the wire
- For a cube, all the sides are of equal lengths
- Let the length of a side be *L*
  - So, the volume of the cube is $V = L^3$
  - The cross-sectional area in contact with the plates is equal to $A = L^2$

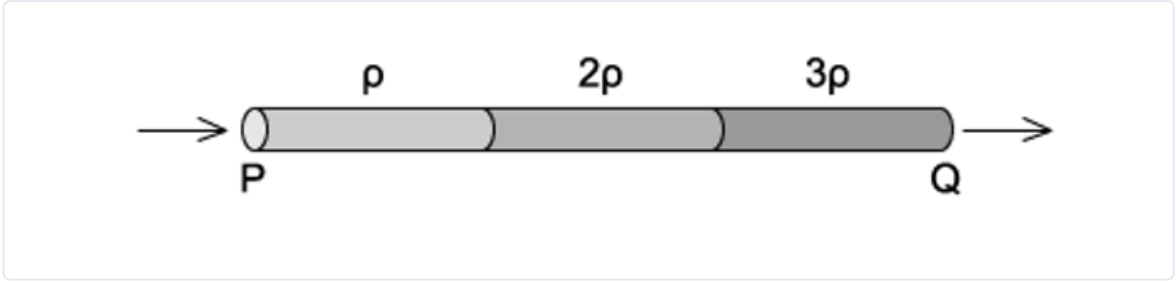
- Resistance is therefore equal to $R = \frac{\rho L}{L^2} = \frac{\rho}{L}$
- But since $V = L^3$, length *L* is equal to $L = \sqrt[3]{V} = V^{\frac{1}{3}}$
    - So, resistance $R = \frac{\rho}{V^{\frac{1}{3}}}$

Exam Tip: An alternative (and potentially quicker) method is to consider the units of the quantities involved, only choice **C** gives the unit of resistance (Ω):

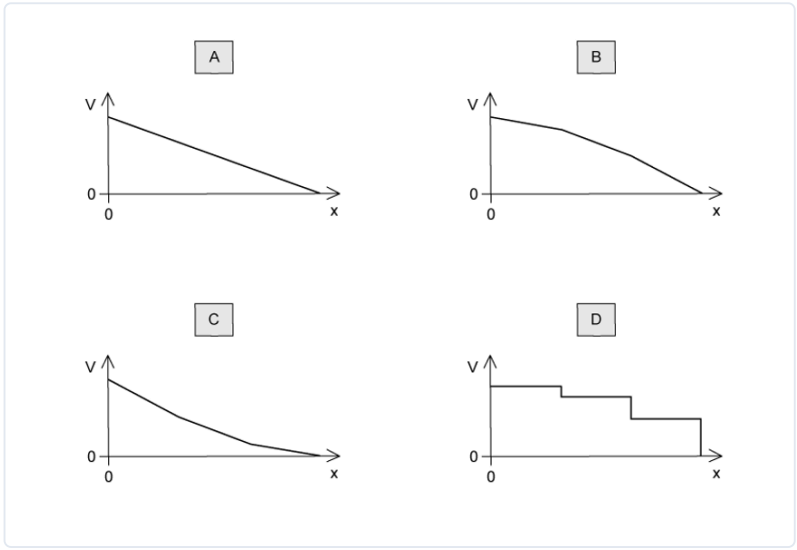
- Unit of resistivity = Ωm
- Unit of volume V = m^3
- Unit of $\frac{\rho}{V^{\frac{1}{3}}} = \Omega m m^{3\,-\frac{1}{3}} = \Omega m \quad m^{-1} = \Omega$

Question 55 Hard [1 mark]

A wire PQ is made of three different materials, with resistivities ρ, 2ρ and 3ρ. There is a current I in this composite wire, as shown.



Which graph best shows how the potential V along the wire varies with distance x from P ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Ohm's law tells us potential difference is equal to $V = IR$
- The resistance R of a wire is proportional to the resistivity ρ of the material from which the wire is made $R = \frac{\rho L}{A}$
- When the resistivity of the material is greater, its resistance is also greater and therefore, the potential difference across it is greater
    - So, the potential difference is greatest across the 3ρ material

    - On the graph, this will be represented by three separate lines with increasing steepness
- Note that the graph gives the potential V along the wire on the y-axis while the *V* from represents the potential difference – which is the change in potential
- The change in potential ΔV as the distance changes Δx is represented by the gradient, which is constant for each value of ρ
- Resistance of the wire also depends on the length of wire being considered. So, as x increases, the length of wire being considered increases and so, as *x* increases, the difference in potential *V* at that particular value of *x* is greater
    - This means in general, as *x* increases, *V* decreases, even if the resistivity is constant
- Therefore, the only graph matching this description is option B

A is incorrect because the gradient is constant everywhere

C is incorrect because the gradient of the straight lines decreases as *x* increases

D is incorrect because the gradient is zero

Question 56 Hard [1 mark]

A strain gauge consists of a length of wire with a uniform cross-sectional area. Its resistance is 2.50 kΩ. It is attached to a large pipe.

When the pipe expands, the strain gauge changes in dimensions, such that the length increases by 4% and the diameter reduced by 1%.

What is the new resistance of the strain gauge?

A 2.60 kΩ

B 2.65 kΩ

C 2.80 kΩ

D 3.00 kΩ

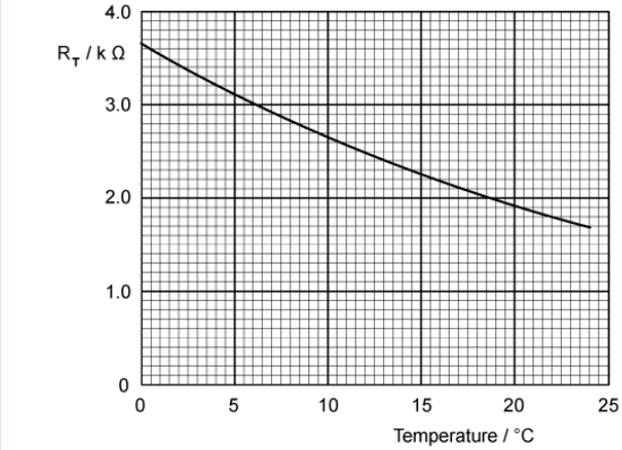
Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Cross-sectional area of a wire is equal to $A = \pi r^2 = \pi \frac{d}{2}^2 = \pi \frac{d}{2}^2 = \frac{\pi d^2}{4}$
- When the pipe expands
  - The new length is given by $L' = 1 + 0.04L = 1.04L$
  - The new cross-sectional area is given by $A' = 1 - 0.01^2 A = 0.99^2 A$ (squared because A is proportional to d^2)
  - The resistivity ρ doesn't change since the material stays the same
- So, the new resistance is equal to
  - $R' = \frac{\rho L'}{A'} = \frac{1.04 \rho L}{0.99^2 A} = \frac{1.04}{0.99^2} R = \frac{1.04}{0.99^2} \times 2.5 = 2.653 \quad \text{k}\Omega$
- Therefore, the new resistance is 2.65 kΩ

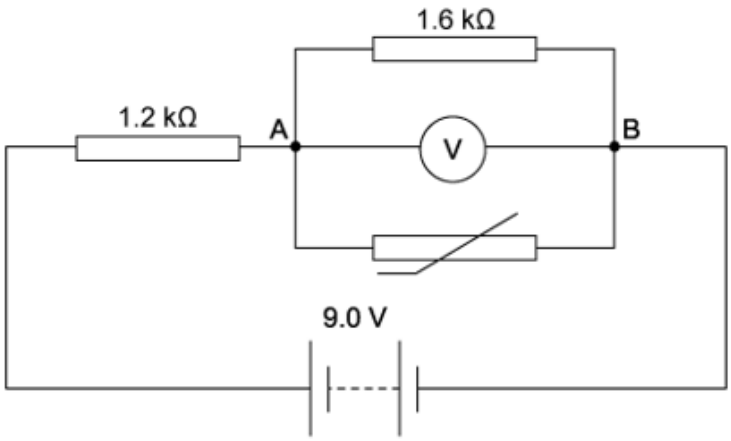
Question 57 Hard [1 mark]

The variation with temperature of the resistance R_T of a thermistor is shown on the graph.



The thermistor is connected into the circuit shown below.

The battery has an e. m .f. of 9.0 V and negligible internal resistance. The voltmeter has infinite resistance.



What is the temperature of the thermistor when the reading on the voltmeter is 4.0 V?

A 2.5 °C

B 13.0 °C

C 16.0 °C

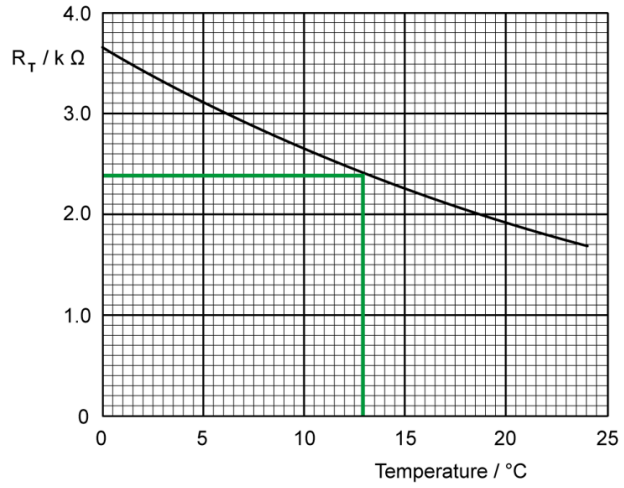
D

24.0 °C

Mark Scheme & Explanation:

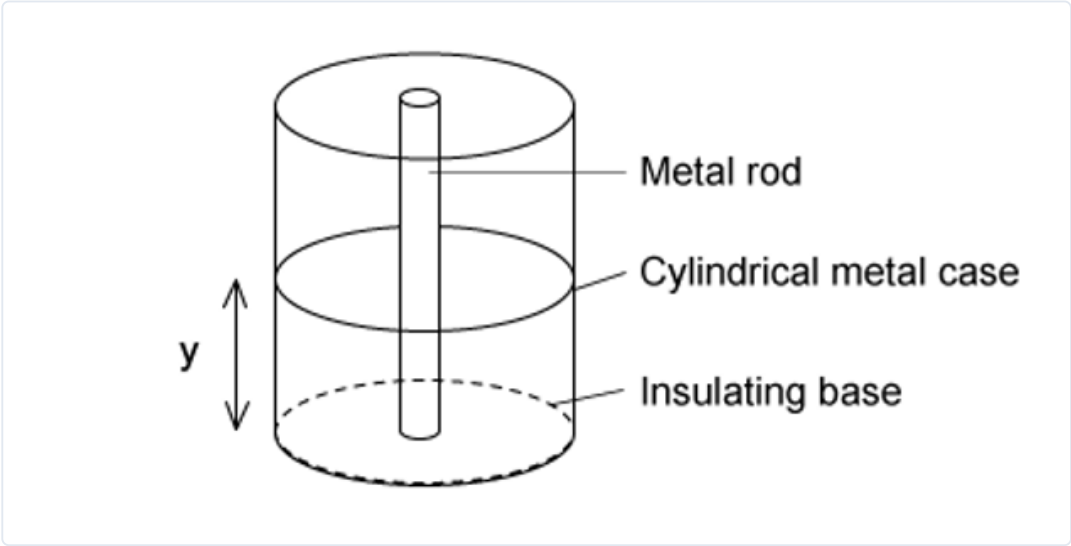
*The correct answer is **B** because:*

- Ohm's law states $V = IR$, V is directly proportional to resistance R
- We are told the reading on the voltmeter is = 4.0 V
- So, the p.d. across the 1.2 kΩ is 9 – 4 = 5.0 V
- A p. d. of 5.0 V corresponds to 1.2 kΩ
  - So, 4.0 V would correspond to a resistance of $\frac{4.0V}{5.0V} \times 1200\ \Omega$
- The total resistance between A and B $= \frac{4}{5} \times 1200 = 960\ \Omega$
- The thermistor and 1.6 kΩ resistor are in parallel so
  - $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$
  - $\frac{1}{960} = \frac{1}{1600} + \frac{1}{R_T}$
  - $\frac{1}{R_T} = \frac{1}{2400}$
- Therefore, the resistance of the thermistor is 2.4 kΩ
- From the graph, the temperature that corresponds to 2.4 kΩ is 11.0 °C



Question 58 Hard [1 mark]

A conducting liquid fills a cylinder metal case to a depth *y* as shown in the diagram.



The resistance between the case and the metal rod is

A proportional to *y*

B proportional to y^2

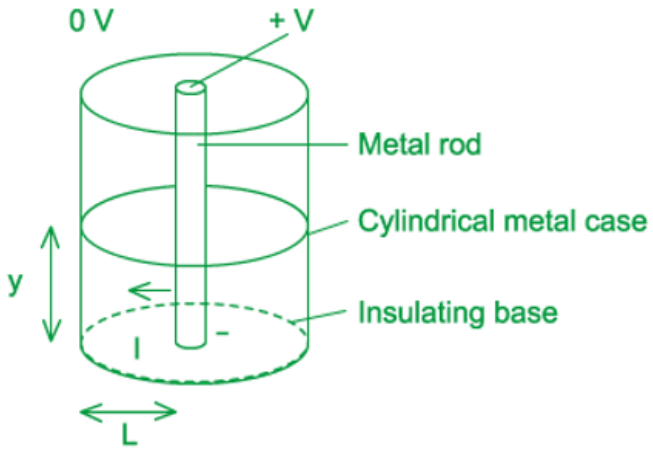
C inversely proportional to *y*

D inversely proportional to y^2

Mark Scheme & Explanation:

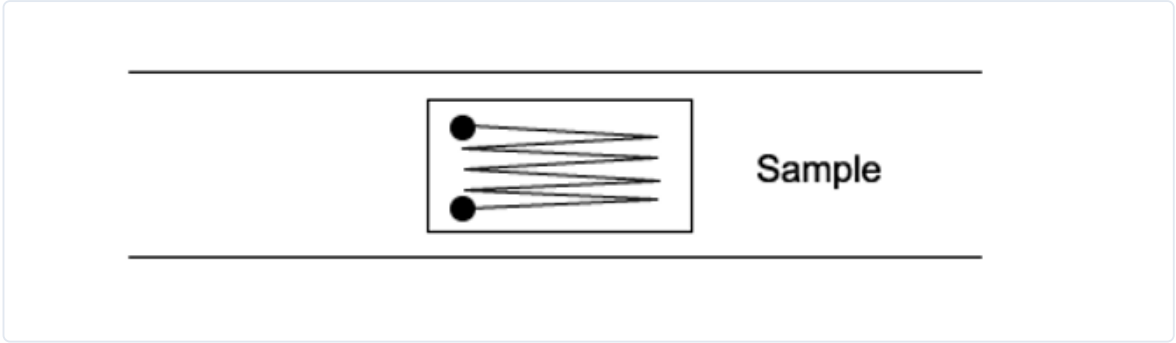
The correct answer is ***C*** *because:*

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- When a potential difference is applied across the metal rod and the case, current *I* will flow through the liquid as shown
- The current will flow over a length *L* and through a cross-sectional area *A*
- In this case, the cross-sectional area is the surface area of the cylinder $2\pi rh$, where *h* is equal to the depth *y*
- So, the cross-sectional area *A* is proportional to *y*
    - Therefore, *R* is inversely proportional to *y*


0 V
+ V
Metal rod
Cylindrical metal case
y
Insulating base
I
L

Question 59 Hard [1 mark]

The following strain gauge consists of a fine metal wire, so that an applied tension on the strain gauge results in changes in the length and cross-sectional area of the wire.



When the length of the wire is stretched by 5%, what is the percentage change in the resistance of the wire? Assume that the volume of a wire remains constant when it is stretched.

A + 5%

B – 5%

C + 10%

D – 10%

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Let's call the new length and cross-sectional area L' and A' respectively
- Since the volume remains constant
  - $V = AL = A'L'$
- When the wire expands
  - The new length is given by $L' = 1 + 0.05L = 1.05L$
  - The new cross-sectional area is given by $V = AL = 1.05L\frac{A}{1.05}$
  - So, $A' = \frac{A}{1.05}$
  - The resistivity ρ doesn't change since the material stays the same
- So, the new resistance is equal to

- $R' = \frac{\rho L'}{A'} = \rho 1.05 \frac{1.05}{A} = 1.05^2 \frac{\rho L}{A} = 1.10R$

- Therefore, the percentage change in *R* is equal to
  - (1.10 – 1.00) × 100 % = 10 %

Question 60 Hard [1 mark]

Aluminum and copper rods are designed to have the same length and the same resistance.

The resistivity of copper is half that of aluminum, but its density is three times that of aluminum.

Determine the ratio of the mass of the aluminum rod to the mass of the copper rod.

A $\frac{1}{6}$

B $\frac{2}{3}$

C $\frac{3}{2}$

D $\frac{6}{1}$

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- In this question we will refer to density by *d* and resistivity by *ρ*
- Density of a material is given by $d = \frac{m}{v}$
  - The mass of the aluminium rod is $m_A = d_A V_A$
  - The mass of the copper rod is $m_c = d_c V_c$
- The volume of a rod is given by $V = AL$
  - The volume of the aluminium rod is $V_A = A_A L$
  - The volume of the copper rod is $V_c = A_c L$
- The ratio of mass of aluminium to mass of copper is equal to
  - $\frac{m_A}{m_c} = \frac{d_A V_A}{d_c V_c} = \frac{d_A A_A L}{d_c A_c L} = \frac{d_A A_A}{d_c A_c}$
- Since $d_c = 3d_A$ (the density of copper is three times that of aluminium)
  - $\frac{m_A}{m_c} = \frac{d_A \ A_A}{3d_A A_c} = \frac{1}{3}\frac{A_A}{A_c}$
- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Since the resistance and length of the rods are the same, $\frac{\rho}{A}$ is constant, so:
  - $\frac{\rho_c}{A_c} = \frac{\rho_A}{A_A}$
  - $\frac{A_A}{A_c} = \frac{\rho_A}{\rho_c}$
- Since $\rho_A = 2\rho_c$ (the resistivity of copper is half that of aluminium)

- $\frac{A_A}{A_c} = \frac{\rho_A}{\rho_C} = \frac{2}{1}$

- Therefore, the ratio $\frac{mass\ \ of\ \ aluminium}{mass\ \ of\ \ copper} = \frac{m_A}{m_c} = \frac{1}{3}\frac{A_A}{A_c} = \frac{1}{3} \times \frac{2}{1} = \frac{2}{3}$